

脚本	主要用途/训练策略	模型结构差异	备注
♦ train_pre.py	baseline 单域训练	PathConv + CNN + MLP	基线脚本
♦ train_balanced_sampling.py	平衡采样、跨工艺预训练/微调	PathConv + CNN + MLP	7/130 训练流程
♦ train_base_pt_ft.py	预训练+微调	PathConv + CNN + MLP	与 baseline 类似
♦ train_balanced_sampling_sep_mlp.py	平衡采样 + 双 MLP	SepPathModel (7nm/130nm 双 MLP)	仍是 path/CNN
♦ train_balanced_sampling_sep_mlp_disentangle.py	双 MLP + 特征解耦	DisSepPathModel (node/others 拆分)	解耦版
♦ train_balanced_sampling_sep_mlp_bayesian.py	双 MLP + 贝叶斯	SepBayesModel (MC Dropout)	不含解耦
♦ train_balanced_sampling_sep_mlp_bayesian_res.py	双 MLP + 贝叶斯 + 残差	SepBayesModel + MlpBayesRes	残差系数可配
♦ train_balanced_sampling_sep_mlp_disentangle_bayesian_res.py	解耦 + 贝叶斯 + 残差	SepBayesDisentangleModel	组合版

图 1:

1 代码运行结果

1.1 train.py

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-----Start training-----
[Ep   5] TrainLoss 0.947607 | ValLoss 0.935506 | ValR2 0.0784
[Ep  10] TrainLoss 0.823693 | ValLoss 0.809458 | ValR2 0.2026
[Ep  15] TrainLoss 0.680557 | ValLoss 0.663603 | ValR2 0.3463
[Ep  20] TrainLoss 0.513411 | ValLoss 0.497920 | ValR2 0.5095
[Ep  25] TrainLoss 0.343258 | ValLoss 0.336665 | ValR2 0.6683
[Ep  30] TrainLoss 0.224037 | ValLoss 0.234793 | ValR2 0.7687
[Ep  35] TrainLoss 0.180900 | ValLoss 0.215747 | ValR2 0.7875
[Ep  40] TrainLoss 0.146404 | ValLoss 0.217511 | ValR2 0.7857
[Ep  45] TrainLoss 0.105966 | ValLoss 0.214824 | ValR2 0.7884
[Ep  50] TrainLoss 0.078862 | ValLoss 0.192982 | ValR2 0.8099
[Ep  55] TrainLoss 0.061591 | ValLoss 0.174721 | ValR2 0.8279
[Ep  60] TrainLoss 0.049539 | ValLoss 0.172322 | ValR2 0.8302
[Ep  65] TrainLoss 0.039761 | ValLoss 0.185246 | ValR2 0.8175
[Ep  70] TrainLoss 0.032694 | ValLoss 0.205578 | ValR2 0.7975
[Ep  75] TrainLoss 0.028391 | ValLoss 0.218205 | ValR2 0.7850
[Ep  80] TrainLoss 0.024710 | ValLoss 0.218596 | ValR2 0.7847
[Ep  85] TrainLoss 0.022219 | ValLoss 0.218986 | ValR2 0.7843
[Ep  90] TrainLoss 0.020316 | ValLoss 0.224725 | ValR2 0.7786
[Ep  95] TrainLoss 0.018739 | ValLoss 0.232561 | ValR2 0.7709
[Ep 100] TrainLoss 0.017409 | ValLoss 0.236332 | ValR2 0.7672
[Ep 105] TrainLoss 0.016232 | ValLoss 0.238136 | ValR2 0.7654
[Ep 110] TrainLoss 0.015120 | ValLoss 0.241249 | ValR2 0.7623
[Ep 115] TrainLoss 0.014055 | ValLoss 0.244693 | ValR2 0.7589

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[Ep 120]	TrainLoss	0.013062		ValLoss	0.247574		ValR2	0.7561
[Ep 125]	TrainLoss	0.012172		ValLoss	0.249981		ValR2	0.7537
[Ep 130]	TrainLoss	0.011312		ValLoss	0.252702		ValR2	0.7511
[Ep 135]	TrainLoss	0.010417		ValLoss	0.255230		ValR2	0.7486
[Ep 140]	TrainLoss	0.009547		ValLoss	0.256906		ValR2	0.7469
[Ep 145]	TrainLoss	0.008761		ValLoss	0.257304		ValR2	0.7465
[Ep 150]	TrainLoss	0.008051		ValLoss	0.258640		ValR2	0.7452
[Ep 155]	TrainLoss	0.007411		ValLoss	0.261275		ValR2	0.7426
[Ep 160]	TrainLoss	0.006825		ValLoss	0.263033		ValR2	0.7409
[Ep 165]	TrainLoss	0.006276		ValLoss	0.263993		ValR2	0.7399
[Ep 170]	TrainLoss	0.005786		ValLoss	0.264290		ValR2	0.7396
[Ep 175]	TrainLoss	0.005340		ValLoss	0.264648		ValR2	0.7393
[Ep 180]	TrainLoss	0.004951		ValLoss	0.264745		ValR2	0.7392
[Ep 185]	TrainLoss	0.004607		ValLoss	0.264935		ValR2	0.7390
[Ep 190]	TrainLoss	0.004301		ValLoss	0.265220		ValR2	0.7387
[Ep 195]	TrainLoss	0.004024		ValLoss	0.265457		ValR2	0.7385
[Ep 200]	TrainLoss	0.003779		ValLoss	0.265442		ValR2	0.7385
[Ep 205]	TrainLoss	0.003565		ValLoss	0.265339		ValR2	0.7386
[Ep 210]	TrainLoss	0.003371		ValLoss	0.265798		ValR2	0.7382
[Ep 215]	TrainLoss	0.003199		ValLoss	0.266347		ValR2	0.7376
[Ep 220]	TrainLoss	0.003045		ValLoss	0.266622		ValR2	0.7373
[Ep 225]	TrainLoss	0.002905		ValLoss	0.266833		ValR2	0.7371
[Ep 230]	TrainLoss	0.002779		ValLoss	0.267086		ValR2	0.7369
[Ep 235]	TrainLoss	0.002664		ValLoss	0.267182		ValR2	0.7368
[Ep 240]	TrainLoss	0.002562		ValLoss	0.267387		ValR2	0.7366
[Ep 245]	TrainLoss	0.002472		ValLoss	0.267810		ValR2	0.7362
[Ep 250]	TrainLoss	0.002387		ValLoss	0.268330		ValR2	0.7357
[Ep 255]	TrainLoss	0.002298		ValLoss	0.268899		ValR2	0.7351
[Ep 260]	TrainLoss	0.002215		ValLoss	0.268930		ValR2	0.7351
[Ep 265]	TrainLoss	0.002142		ValLoss	0.269004		ValR2	0.7350
[Ep 270]	TrainLoss	0.002078		ValLoss	0.269121		ValR2	0.7349
[Ep 275]	TrainLoss	0.002016		ValLoss	0.269186		ValR2	0.7348
[Ep 280]	TrainLoss	0.001960		ValLoss	0.268921		ValR2	0.7351
[Ep 285]	TrainLoss	0.001910		ValLoss	0.268645		ValR2	0.7354
[Ep 290]	TrainLoss	0.001866		ValLoss	0.268450		ValR2	0.7355
[Ep 295]	TrainLoss	0.001828		ValLoss	0.268403		ValR2	0.7356
[Ep 300]	TrainLoss	0.001796		ValLoss	0.268414		ValR2	0.7356
[Ep 305]	TrainLoss	0.001767		ValLoss	0.268688		ValR2	0.7353
[Ep 310]	TrainLoss	0.001741		ValLoss	0.268971		ValR2	0.7350

[Ep 315]	TrainLoss	0.001716		ValLoss	0.269487		ValR2	0.7345
[Ep 320]	TrainLoss	0.001694		ValLoss	0.270191		ValR2	0.7338
[Ep 325]	TrainLoss	0.001674		ValLoss	0.270784		ValR2	0.7332
[Ep 330]	TrainLoss	0.001655		ValLoss	0.271325		ValR2	0.7327
[Ep 335]	TrainLoss	0.001637		ValLoss	0.271681		ValR2	0.7324
[Ep 340]	TrainLoss	0.001619		ValLoss	0.272103		ValR2	0.7319
[Ep 345]	TrainLoss	0.001602		ValLoss	0.272528		ValR2	0.7315
[Ep 350]	TrainLoss	0.001586		ValLoss	0.272900		ValR2	0.7312
[Ep 355]	TrainLoss	0.001570		ValLoss	0.273229		ValR2	0.7308
[Ep 360]	TrainLoss	0.001555		ValLoss	0.273409		ValR2	0.7307
[Ep 365]	TrainLoss	0.001541		ValLoss	0.273664		ValR2	0.7304
[Ep 370]	TrainLoss	0.001528		ValLoss	0.273955		ValR2	0.7301
[Ep 375]	TrainLoss	0.001515		ValLoss	0.274252		ValR2	0.7298
[Ep 380]	TrainLoss	0.001503		ValLoss	0.274587		ValR2	0.7295
[Ep 385]	TrainLoss	0.001492		ValLoss	0.274873		ValR2	0.7292
[Ep 390]	TrainLoss	0.001481		ValLoss	0.275194		ValR2	0.7289
[Ep 395]	TrainLoss	0.001471		ValLoss	0.275528		ValR2	0.7286
[Ep 400]	TrainLoss	0.001461		ValLoss	0.275840		ValR2	0.7283
[Ep 405]	TrainLoss	0.001451		ValLoss	0.276175		ValR2	0.7279
[Ep 410]	TrainLoss	0.001442		ValLoss	0.276527		ValR2	0.7276
[Ep 415]	TrainLoss	0.001433		ValLoss	0.276771		ValR2	0.7273
[Ep 420]	TrainLoss	0.001424		ValLoss	0.277093		ValR2	0.7270
[Ep 425]	TrainLoss	0.001416		ValLoss	0.277389		ValR2	0.7267
[Ep 430]	TrainLoss	0.001409		ValLoss	0.277667		ValR2	0.7265
[Ep 435]	TrainLoss	0.001401		ValLoss	0.278016		ValR2	0.7261
[Ep 440]	TrainLoss	0.001394		ValLoss	0.278337		ValR2	0.7258
[Ep 445]	TrainLoss	0.001387		ValLoss	0.278629		ValR2	0.7255
[Ep 450]	TrainLoss	0.001380		ValLoss	0.278880		ValR2	0.7253
[Ep 455]	TrainLoss	0.001373		ValLoss	0.279130		ValR2	0.7250
[Ep 460]	TrainLoss	0.001366		ValLoss	0.279366		ValR2	0.7248
[Ep 465]	TrainLoss	0.001360		ValLoss	0.279658		ValR2	0.7245
[Ep 470]	TrainLoss	0.001354		ValLoss	0.279977		ValR2	0.7242
[Ep 475]	TrainLoss	0.001348		ValLoss	0.280286		ValR2	0.7239
[Ep 480]	TrainLoss	0.001342		ValLoss	0.280574		ValR2	0.7236
[Ep 485]	TrainLoss	0.001337		ValLoss	0.280827		ValR2	0.7234
[Ep 490]	TrainLoss	0.001331		ValLoss	0.281060		ValR2	0.7231
[Ep 495]	TrainLoss	0.001326		ValLoss	0.281306		ValR2	0.7229
[Ep 500]	TrainLoss	0.001320		ValLoss	0.281521		ValR2	0.7227
[Ep 505]	TrainLoss	0.001315		ValLoss	0.281713		ValR2	0.7225

[Ep 510]	TrainLoss	0.001310		ValLoss	0.281904		ValR2	0.7223
[Ep 515]	TrainLoss	0.001305		ValLoss	0.282129		ValR2	0.7221
[Ep 520]	TrainLoss	0.001300		ValLoss	0.282387		ValR2	0.7218
[Ep 525]	TrainLoss	0.001296		ValLoss	0.282613		ValR2	0.7216
[Ep 530]	TrainLoss	0.001292		ValLoss	0.282777		ValR2	0.7214
[Ep 535]	TrainLoss	0.001287		ValLoss	0.282931		ValR2	0.7213
[Ep 540]	TrainLoss	0.001283		ValLoss	0.283128		ValR2	0.7211
[Ep 545]	TrainLoss	0.001279		ValLoss	0.283339		ValR2	0.7209
[Ep 550]	TrainLoss	0.001275		ValLoss	0.283576		ValR2	0.7206
[Ep 555]	TrainLoss	0.001271		ValLoss	0.283762		ValR2	0.7205
[Ep 560]	TrainLoss	0.001267		ValLoss	0.283929		ValR2	0.7203
[Ep 565]	TrainLoss	0.001263		ValLoss	0.284078		ValR2	0.7201
[Ep 570]	TrainLoss	0.001260		ValLoss	0.284225		ValR2	0.7200
[Ep 575]	TrainLoss	0.001257		ValLoss	0.284403		ValR2	0.7198
[Ep 580]	TrainLoss	0.001253		ValLoss	0.284578		ValR2	0.7197
[Ep 585]	TrainLoss	0.001250		ValLoss	0.284782		ValR2	0.7195
[Ep 590]	TrainLoss	0.001247		ValLoss	0.285010		ValR2	0.7192
[Ep 595]	TrainLoss	0.001244		ValLoss	0.285146		ValR2	0.7191
[Ep 600]	TrainLoss	0.001242		ValLoss	0.285265		ValR2	0.7190
[Ep 605]	TrainLoss	0.001239		ValLoss	0.285410		ValR2	0.7188
[Ep 610]	TrainLoss	0.001236		ValLoss	0.285566		ValR2	0.7187
[Ep 615]	TrainLoss	0.001234		ValLoss	0.285745		ValR2	0.7185
[Ep 620]	TrainLoss	0.001231		ValLoss	0.285886		ValR2	0.7184
[Ep 625]	TrainLoss	0.001229		ValLoss	0.286019		ValR2	0.7182
[Ep 630]	TrainLoss	0.001226		ValLoss	0.286166		ValR2	0.7181
[Ep 635]	TrainLoss	0.001224		ValLoss	0.286336		ValR2	0.7179
[Ep 640]	TrainLoss	0.001221		ValLoss	0.286501		ValR2	0.7178
[Ep 645]	TrainLoss	0.001219		ValLoss	0.286701		ValR2	0.7176
[Ep 650]	TrainLoss	0.001217		ValLoss	0.286826		ValR2	0.7174
[Ep 655]	TrainLoss	0.001214		ValLoss	0.287002		ValR2	0.7173
[Ep 660]	TrainLoss	0.001212		ValLoss	0.287160		ValR2	0.7171
[Ep 665]	TrainLoss	0.001210		ValLoss	0.287301		ValR2	0.7170
[Ep 670]	TrainLoss	0.001208		ValLoss	0.287434		ValR2	0.7168
[Ep 675]	TrainLoss	0.001206		ValLoss	0.287557		ValR2	0.7167
[Ep 680]	TrainLoss	0.001204		ValLoss	0.287686		ValR2	0.7166
[Ep 685]	TrainLoss	0.001202		ValLoss	0.287822		ValR2	0.7165
[Ep 690]	TrainLoss	0.001200		ValLoss	0.287954		ValR2	0.7163
[Ep 695]	TrainLoss	0.001198		ValLoss	0.288078		ValR2	0.7162
[Ep 700]	TrainLoss	0.001196		ValLoss	0.288212		ValR2	0.7161

[Ep 705]	TrainLoss	0.001195		ValLoss	0.288338		ValR2	0.7160
[Ep 710]	TrainLoss	0.001193		ValLoss	0.288481		ValR2	0.7158
[Ep 715]	TrainLoss	0.001191		ValLoss	0.288614		ValR2	0.7157
[Ep 720]	TrainLoss	0.001190		ValLoss	0.288742		ValR2	0.7156
[Ep 725]	TrainLoss	0.001188		ValLoss	0.288840		ValR2	0.7155
[Ep 730]	TrainLoss	0.001187		ValLoss	0.288934		ValR2	0.7154
[Ep 735]	TrainLoss	0.001185		ValLoss	0.289041		ValR2	0.7153
[Ep 740]	TrainLoss	0.001183		ValLoss	0.289177		ValR2	0.7151
[Ep 745]	TrainLoss	0.001182		ValLoss	0.289248		ValR2	0.7151
[Ep 750]	TrainLoss	0.001180		ValLoss	0.289357		ValR2	0.7149
[Ep 755]	TrainLoss	0.001179		ValLoss	0.289446		ValR2	0.7149
[Ep 760]	TrainLoss	0.001177		ValLoss	0.289515		ValR2	0.7148
[Ep 765]	TrainLoss	0.001176		ValLoss	0.289595		ValR2	0.7147
[Ep 770]	TrainLoss	0.001174		ValLoss	0.289658		ValR2	0.7146
[Ep 775]	TrainLoss	0.001173		ValLoss	0.289702		ValR2	0.7146
[Ep 780]	TrainLoss	0.001172		ValLoss	0.289744		ValR2	0.7146
[Ep 785]	TrainLoss	0.001170		ValLoss	0.289749		ValR2	0.7146
[Ep 790]	TrainLoss	0.001169		ValLoss	0.289774		ValR2	0.7145
[Ep 795]	TrainLoss	0.001168		ValLoss	0.289774		ValR2	0.7145
[Ep 800]	TrainLoss	0.001166		ValLoss	0.289755		ValR2	0.7146
[Ep 805]	TrainLoss	0.001165		ValLoss	0.289773		ValR2	0.7145
[Ep 810]	TrainLoss	0.001164		ValLoss	0.289799		ValR2	0.7145
[Ep 815]	TrainLoss	0.001163		ValLoss	0.289775		ValR2	0.7145
[Ep 820]	TrainLoss	0.001161		ValLoss	0.289752		ValR2	0.7146
[Ep 825]	TrainLoss	0.001160		ValLoss	0.289752		ValR2	0.7146
[Ep 830]	TrainLoss	0.001159		ValLoss	0.289749		ValR2	0.7146
[Ep 835]	TrainLoss	0.001158		ValLoss	0.289796		ValR2	0.7145
[Ep 840]	TrainLoss	0.001157		ValLoss	0.289795		ValR2	0.7145
[Ep 845]	TrainLoss	0.001156		ValLoss	0.289789		ValR2	0.7145
[Ep 850]	TrainLoss	0.001155		ValLoss	0.289786		ValR2	0.7145
[Ep 855]	TrainLoss	0.001154		ValLoss	0.289797		ValR2	0.7145
[Ep 860]	TrainLoss	0.001153		ValLoss	0.289766		ValR2	0.7145
[Ep 865]	TrainLoss	0.001152		ValLoss	0.289716		ValR2	0.7146
[Ep 870]	TrainLoss	0.001151		ValLoss	0.289731		ValR2	0.7146
[Ep 875]	TrainLoss	0.001150		ValLoss	0.289695		ValR2	0.7146
[Ep 880]	TrainLoss	0.001149		ValLoss	0.289667		ValR2	0.7146
[Ep 885]	TrainLoss	0.001148		ValLoss	0.289651		ValR2	0.7147
[Ep 890]	TrainLoss	0.001147		ValLoss	0.289583		ValR2	0.7147
[Ep 895]	TrainLoss	0.001146		ValLoss	0.289505		ValR2	0.7148

[Ep 900]	TrainLoss	0.001145		ValLoss	0.289474		ValR2	0.7148
[Ep 905]	TrainLoss	0.001144		ValLoss	0.289397		ValR2	0.7149
[Ep 910]	TrainLoss	0.001143		ValLoss	0.289322		ValR2	0.7150
[Ep 915]	TrainLoss	0.001142		ValLoss	0.289259		ValR2	0.7150
[Ep 920]	TrainLoss	0.001141		ValLoss	0.289207		ValR2	0.7151
[Ep 925]	TrainLoss	0.001140		ValLoss	0.289190		ValR2	0.7151
[Ep 930]	TrainLoss	0.001139		ValLoss	0.289179		ValR2	0.7151
[Ep 935]	TrainLoss	0.001138		ValLoss	0.289161		ValR2	0.7151
[Ep 940]	TrainLoss	0.001137		ValLoss	0.289201		ValR2	0.7151
[Ep 945]	TrainLoss	0.001136		ValLoss	0.289195		ValR2	0.7151
[Ep 950]	TrainLoss	0.001135		ValLoss	0.289189		ValR2	0.7151
[Ep 955]	TrainLoss	0.001134		ValLoss	0.289231		ValR2	0.7151
[Ep 960]	TrainLoss	0.001133		ValLoss	0.289219		ValR2	0.7151
[Ep 965]	TrainLoss	0.001132		ValLoss	0.289221		ValR2	0.7151
[Ep 970]	TrainLoss	0.001131		ValLoss	0.289229		ValR2	0.7151
[Ep 975]	TrainLoss	0.001130		ValLoss	0.289232		ValR2	0.7151
[Ep 980]	TrainLoss	0.001129		ValLoss	0.289223		ValR2	0.7151
[Ep 985]	TrainLoss	0.001129		ValLoss	0.289191		ValR2	0.7151
[Ep 990]	TrainLoss	0.001128		ValLoss	0.289172		ValR2	0.7151
[Ep 995]	TrainLoss	0.001127		ValLoss	0.289123		ValR2	0.7152
[Ep 1000]	TrainLoss	0.001126		ValLoss	0.289121		ValR2	0.7152

1.2 train_base_pt_ft.py

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Start training
[Pretrain] e0, train_loss:0.5202, r2:0.004, val_loss:0.9405, val_r2:0.073
Model successfully saved
[Pretrain] e1, train_loss:0.4079, r2:0.219, val_loss:0.8297, val_r2:0.183
Model successfully saved
[Pretrain] e2, train_loss:0.3209, r2:0.385, val_loss:0.7342, val_r2:0.277
Model successfully saved
[Pretrain] e3, train_loss:0.2781, r2:0.467, val_loss:0.6445, val_r2:0.365
Model successfully saved
[Pretrain] e4, train_loss:0.2674, r2:0.488, val_loss:0.6192, val_r2:0.390
Model successfully saved
[Pretrain] e5, train_loss:0.2511, r2:0.519, val_loss:0.6387, val_r2:0.371
[Pretrain] e6, train_loss:0.2414, r2:0.538, val_loss:0.6041, val_r2:0.405
Model successfully saved
[Pretrain] e7, train_loss:0.2304, r2:0.559, val_loss:0.5980, val_r2:0.411

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Model successfully saved

[Pretrain] e8, train_loss:0.2237, r2:0.572, val_loss:0.6059, val_r2:0.403
[Pretrain] e9, train_loss:0.2169, r2:0.585, val_loss:0.5932, val_r2:0.416

Model successfully saved

[Pretrain] e10, train_loss:0.2129, r2:0.592, val_loss:0.6230, val_r2:0.386
[Pretrain] e11, train_loss:0.2087, r2:0.600, val_loss:0.6330, val_r2:0.376
[Pretrain] e12, train_loss:0.2053, r2:0.607, val_loss:0.6556, val_r2:0.354
[Pretrain] e13, train_loss:0.2032, r2:0.611, val_loss:0.6839, val_r2:0.326
[Pretrain] e14, train_loss:0.2009, r2:0.615, val_loss:0.7462, val_r2:0.265
[Pretrain] e15, train_loss:0.2010, r2:0.615, val_loss:0.7415, val_r2:0.270
[Pretrain] e16, train_loss:0.1987, r2:0.620, val_loss:0.7597, val_r2:0.252
[Pretrain] e17, train_loss:0.1973, r2:0.622, val_loss:0.7802, val_r2:0.231
[Pretrain] e18, train_loss:0.1956, r2:0.625, val_loss:0.8246, val_r2:0.188
[Pretrain] e19, train_loss:0.1935, r2:0.629, val_loss:0.8530, val_r2:0.160
[Pretrain] e20, train_loss:0.1925, r2:0.631, val_loss:0.8771, val_r2:0.136
[Pretrain] e21, train_loss:0.1912, r2:0.634, val_loss:0.8851, val_r2:0.128
[Pretrain] e22, train_loss:0.1910, r2:0.634, val_loss:0.9311, val_r2:0.083
[Pretrain] e23, train_loss:0.1897, r2:0.637, val_loss:0.9873, val_r2:0.027
[Pretrain] e24, train_loss:0.1880, r2:0.640, val_loss:1.0235, val_r2:-0.008
[Pretrain] e25, train_loss:0.1873, r2:0.641, val_loss:1.0505, val_r2:-0.035
[Pretrain] e26, train_loss:0.1861, r2:0.644, val_loss:1.0622, val_r2:-0.046
[Pretrain] e27, train_loss:0.1854, r2:0.645, val_loss:1.1040, val_r2:-0.088
[Pretrain] e28, train_loss:0.1851, r2:0.645, val_loss:1.1666, val_r2:-0.149
[Pretrain] e29, train_loss:0.1840, r2:0.648, val_loss:1.1890, val_r2:-0.171
[Pretrain] e30, train_loss:0.1833, r2:0.649, val_loss:1.2001, val_r2:-0.182
[Pretrain] e31, train_loss:0.1829, r2:0.650, val_loss:1.2310, val_r2:-0.213
[Pretrain] e32, train_loss:0.1825, r2:0.650, val_loss:1.2869, val_r2:-0.268
[Pretrain] e33, train_loss:0.1818, r2:0.652, val_loss:1.3151, val_r2:-0.295
[Pretrain] e34, train_loss:0.1813, r2:0.653, val_loss:1.3461, val_r2:-0.326
[Pretrain] e35, train_loss:0.1809, r2:0.653, val_loss:1.3938, val_r2:-0.373
[Pretrain] e36, train_loss:0.1807, r2:0.654, val_loss:1.4139, val_r2:-0.393
[Pretrain] e37, train_loss:0.1815, r2:0.652, val_loss:1.4668, val_r2:-0.445
[Pretrain] e38, train_loss:0.1808, r2:0.654, val_loss:1.5475, val_r2:-0.524
[Pretrain] e39, train_loss:0.1802, r2:0.655, val_loss:1.6076, val_r2:-0.584
[Pretrain] e40, train_loss:0.1812, r2:0.653, val_loss:1.6228, val_r2:-0.599
[Pretrain] e41, train_loss:0.1791, r2:0.657, val_loss:1.6533, val_r2:-0.629
[Pretrain] e42, train_loss:0.1785, r2:0.658, val_loss:1.6719, val_r2:-0.647
[Pretrain] e43, train_loss:0.1785, r2:0.658, val_loss:1.7245, val_r2:-0.699
[Pretrain] e44, train_loss:0.1783, r2:0.659, val_loss:1.7467, val_r2:-0.721

[Pretrain] e45, train_loss:0.1783, r2:0.659, val_loss:1.8107, val_r2:-0.784
[Pretrain] e46, train_loss:0.1786, r2:0.658, val_loss:1.8364, val_r2:-0.809
[Pretrain] e47, train_loss:0.1773, r2:0.660, val_loss:1.8597, val_r2:-0.832
[Pretrain] e48, train_loss:0.1783, r2:0.659, val_loss:1.9152, val_r2:-0.887
[Pretrain] e49, train_loss:0.1764, r2:0.662, val_loss:1.9888, val_r2:-0.959
[Pretrain] e50, train_loss:0.1773, r2:0.660, val_loss:2.0399, val_r2:-1.010
[Pretrain] e51, train_loss:0.1780, r2:0.659, val_loss:2.0937, val_r2:-1.063
[Pretrain] e52, train_loss:0.1774, r2:0.660, val_loss:2.0504, val_r2:-1.020
[Pretrain] e53, train_loss:0.1770, r2:0.661, val_loss:2.1173, val_r2:-1.086
[Pretrain] e54, train_loss:0.1757, r2:0.663, val_loss:2.1309, val_r2:-1.099
[Pretrain] e55, train_loss:0.1754, r2:0.664, val_loss:2.1884, val_r2:-1.156
[Pretrain] e56, train_loss:0.1754, r2:0.664, val_loss:2.2285, val_r2:-1.195
[Pretrain] e57, train_loss:0.1749, r2:0.665, val_loss:2.3195, val_r2:-1.285
[Pretrain] e58, train_loss:0.1748, r2:0.665, val_loss:2.3479, val_r2:-1.313
[Pretrain] e59, train_loss:0.1744, r2:0.666, val_loss:2.3861, val_r2:-1.351
[Pretrain] e60, train_loss:0.1739, r2:0.667, val_loss:2.4465, val_r2:-1.410
[Pretrain] e61, train_loss:0.1739, r2:0.667, val_loss:2.4765, val_r2:-1.440
[Pretrain] e62, train_loss:0.1740, r2:0.667, val_loss:2.5867, val_r2:-1.548
[Pretrain] e63, train_loss:0.1739, r2:0.667, val_loss:2.5937, val_r2:-1.555
[Pretrain] e64, train_loss:0.1735, r2:0.668, val_loss:2.6833, val_r2:-1.643
[Pretrain] e65, train_loss:0.1729, r2:0.669, val_loss:2.7312, val_r2:-1.691
[Pretrain] e66, train_loss:0.1735, r2:0.668, val_loss:2.8107, val_r2:-1.769
[Pretrain] e67, train_loss:0.1741, r2:0.667, val_loss:2.8596, val_r2:-1.817
[Pretrain] e68, train_loss:0.1724, r2:0.670, val_loss:2.8217, val_r2:-1.780
[Pretrain] e69, train_loss:0.1715, r2:0.672, val_loss:2.9141, val_r2:-1.871
[Pretrain] e70, train_loss:0.1714, r2:0.672, val_loss:2.9611, val_r2:-1.917
[Pretrain] e71, train_loss:0.1715, r2:0.672, val_loss:3.0363, val_r2:-1.991
[Pretrain] e72, train_loss:0.1710, r2:0.673, val_loss:3.1434, val_r2:-2.097
[Pretrain] e73, train_loss:0.1701, r2:0.674, val_loss:3.1530, val_r2:-2.106
[Pretrain] e74, train_loss:0.1702, r2:0.674, val_loss:3.3198, val_r2:-2.270
[Pretrain] e75, train_loss:0.1701, r2:0.674, val_loss:3.3145, val_r2:-2.265
[Pretrain] e76, train_loss:0.1703, r2:0.674, val_loss:3.4552, val_r2:-2.404
[Pretrain] e77, train_loss:0.1708, r2:0.673, val_loss:3.4704, val_r2:-2.419
[Pretrain] e78, train_loss:0.1711, r2:0.672, val_loss:3.6226, val_r2:-2.569
[Pretrain] e79, train_loss:0.1707, r2:0.673, val_loss:3.6364, val_r2:-2.582
[Pretrain] e80, train_loss:0.1700, r2:0.674, val_loss:3.7011, val_r2:-2.646
[Pretrain] e81, train_loss:0.1683, r2:0.678, val_loss:3.8090, val_r2:-2.752
[Pretrain] e82, train_loss:0.1703, r2:0.674, val_loss:3.8108, val_r2:-2.754
[Pretrain] e83, train_loss:0.1687, r2:0.677, val_loss:3.9800, val_r2:-2.921

[Pretrain] e84, train_loss:0.1683, r2:0.678, val_loss:3.9835, val_r2:-2.924
 [Pretrain] e85, train_loss:0.1680, r2:0.678, val_loss:4.1540, val_r2:-3.092
 [Pretrain] e86, train_loss:0.1664, r2:0.681, val_loss:4.0888, val_r2:-3.028
 [Pretrain] e87, train_loss:0.1667, r2:0.681, val_loss:4.3342, val_r2:-3.270
 [Pretrain] e88, train_loss:0.1662, r2:0.682, val_loss:4.3225, val_r2:-3.258
 [Pretrain] e89, train_loss:0.1662, r2:0.682, val_loss:4.5995, val_r2:-3.531
 [Pretrain] e90, train_loss:0.1662, r2:0.682, val_loss:4.6131, val_r2:-3.545
 [Pretrain] e91, train_loss:0.1647, r2:0.685, val_loss:4.7780, val_r2:-3.707
 [Pretrain] e92, train_loss:0.1649, r2:0.684, val_loss:4.8458, val_r2:-3.774
 [Pretrain] e93, train_loss:0.1652, r2:0.684, val_loss:4.9361, val_r2:-3.863
 [Pretrain] e94, train_loss:0.1646, r2:0.685, val_loss:5.0723, val_r2:-3.997
 [Pretrain] e95, train_loss:0.1648, r2:0.684, val_loss:5.1869, val_r2:-4.110
 [Pretrain] e96, train_loss:0.1642, r2:0.686, val_loss:5.3122, val_r2:-4.233
 [Pretrain] e97, train_loss:0.1631, r2:0.688, val_loss:5.4680, val_r2:-4.387
 [Pretrain] e98, train_loss:0.1625, r2:0.689, val_loss:5.5936, val_r2:-4.510
 [Pretrain] e99, train_loss:0.1626, r2:0.689, val_loss:5.7604, val_r2:-4.675
 [Pretrain] e100, train_loss:0.1626, r2:0.689, val_loss:5.8501, val_r2:-4.763
 [Pretrain] e101, train_loss:0.1637, r2:0.687, val_loss:6.0782, val_r2:-4.988
 [Pretrain] e102, train_loss:0.1631, r2:0.688, val_loss:6.0714, val_r2:-4.981
 [Pretrain] e103, train_loss:0.1611, r2:0.691, val_loss:6.3008, val_r2:-5.207
 [Pretrain] e104, train_loss:0.1610, r2:0.692, val_loss:6.3610, val_r2:-5.266
 [Pretrain] e105, train_loss:0.1606, r2:0.692, val_loss:6.5476, val_r2:-5.450
 [Pretrain] e106, train_loss:0.1599, r2:0.694, val_loss:6.6837, val_r2:-5.584
 [Pretrain] e107, train_loss:0.1603, r2:0.693, val_loss:6.8388, val_r2:-5.737
 [Pretrain] e108, train_loss:0.1596, r2:0.694, val_loss:7.0554, val_r2:-5.950
 [Pretrain] e109, train_loss:0.1598, r2:0.694, val_loss:7.2111, val_r2:-6.104
 [Pretrain] e110, train_loss:0.1594, r2:0.695, val_loss:7.3375, val_r2:-6.228
 [Pretrain] e111, train_loss:0.1587, r2:0.696, val_loss:7.4943, val_r2:-6.383
 [Pretrain] e112, train_loss:0.1589, r2:0.696, val_loss:7.7117, val_r2:-6.597
 [Pretrain] e113, train_loss:0.1583, r2:0.697, val_loss:7.7435, val_r2:-6.628
 [Pretrain] e114, train_loss:0.1577, r2:0.698, val_loss:8.0089, val_r2:-6.890
 [Pretrain] e115, train_loss:0.1577, r2:0.698, val_loss:8.1292, val_r2:-7.008
 [Pretrain] e116, train_loss:0.1572, r2:0.699, val_loss:8.2019, val_r2:-7.080
 [Pretrain] e117, train_loss:0.1593, r2:0.695, val_loss:8.5326, val_r2:-7.406
 [Pretrain] e118, train_loss:0.1579, r2:0.698, val_loss:8.4602, val_r2:-7.334
 [Pretrain] e119, train_loss:0.1583, r2:0.697, val_loss:8.8727, val_r2:-7.741
 [Pretrain] e120, train_loss:0.1579, r2:0.698, val_loss:8.8692, val_r2:-7.737
 [Pretrain] e121, train_loss:0.1576, r2:0.698, val_loss:8.9445, val_r2:-7.811
 [Pretrain] e122, train_loss:0.1565, r2:0.700, val_loss:9.1993, val_r2:-8.062

[Pretrain] e123, train_loss:0.1569, r2:0.700, val_loss:9.2195, val_r2:-8.082
 [Pretrain] e124, train_loss:0.1569, r2:0.699, val_loss:9.4320, val_r2:-8.292
 [Pretrain] e125, train_loss:0.1560, r2:0.701, val_loss:9.5326, val_r2:-8.391
 [Pretrain] e126, train_loss:0.1551, r2:0.703, val_loss:9.6396, val_r2:-8.496
 [Pretrain] e127, train_loss:0.1552, r2:0.703, val_loss:9.8423, val_r2:-8.696
 [Pretrain] e128, train_loss:0.1544, r2:0.704, val_loss:9.9418, val_r2:-8.794
 [Pretrain] e129, train_loss:0.1531, r2:0.707, val_loss:10.1644, val_r2:-9.013
 [Pretrain] e130, train_loss:0.1543, r2:0.704, val_loss:10.2134, val_r2:-9.062
 [Pretrain] e131, train_loss:0.1546, r2:0.704, val_loss:10.4757, val_r2:-9.320
 [Pretrain] e132, train_loss:0.1547, r2:0.704, val_loss:10.5675, val_r2:-9.410
 [Pretrain] e133, train_loss:0.1531, r2:0.707, val_loss:10.7094, val_r2:-9.550
 [Pretrain] e134, train_loss:0.1543, r2:0.705, val_loss:10.8768, val_r2:-9.715
 [Pretrain] e135, train_loss:0.1521, r2:0.709, val_loss:10.9491, val_r2:-9.786
 [Pretrain] e136, train_loss:0.1516, r2:0.710, val_loss:11.1497, val_r2:-9.984
 [Pretrain] e137, train_loss:0.1513, r2:0.710, val_loss:11.3097, val_r2:-10.14
 [Pretrain] e138, train_loss:0.1520, r2:0.709, val_loss:11.5421, val_r2:-10.37
 [Pretrain] e139, train_loss:0.1521, r2:0.709, val_loss:11.6399, val_r2:-10.46
 [Pretrain] e140, train_loss:0.1508, r2:0.711, val_loss:11.8778, val_r2:-10.70
 [Pretrain] e141, train_loss:0.1505, r2:0.712, val_loss:12.0506, val_r2:-10.87
 [Pretrain] e142, train_loss:0.1503, r2:0.712, val_loss:12.2869, val_r2:-11.10
 [Pretrain] e143, train_loss:0.1514, r2:0.710, val_loss:12.5346, val_r2:-11.34
 [Pretrain] e144, train_loss:0.1522, r2:0.709, val_loss:12.5250, val_r2:-11.33
 [Pretrain] e145, train_loss:0.1546, r2:0.704, val_loss:12.7582, val_r2:-11.56
 [Pretrain] e146, train_loss:0.1525, r2:0.708, val_loss:12.8088, val_r2:-11.61
 [Pretrain] e147, train_loss:0.1505, r2:0.712, val_loss:12.9197, val_r2:-11.72
 [Pretrain] e148, train_loss:0.1495, r2:0.714, val_loss:13.0746, val_r2:-11.88
 [Pretrain] e149, train_loss:0.1489, r2:0.715, val_loss:13.3530, val_r2:-12.15
 [Pretrain] e150, train_loss:0.1491, r2:0.714, val_loss:13.4967, val_r2:-12.29
 [Pretrain] e151, train_loss:0.1486, r2:0.715, val_loss:13.7646, val_r2:-12.56
 [Pretrain] e152, train_loss:0.1486, r2:0.715, val_loss:13.9704, val_r2:-12.76
 [Pretrain] e153, train_loss:0.1490, r2:0.715, val_loss:14.1367, val_r2:-12.92
 [Pretrain] e154, train_loss:0.1497, r2:0.713, val_loss:14.4180, val_r2:-13.20
 [Pretrain] e155, train_loss:0.1478, r2:0.717, val_loss:14.5132, val_r2:-13.29
 [Pretrain] e156, train_loss:0.1481, r2:0.716, val_loss:14.7655, val_r2:-13.54
 [Pretrain] e157, train_loss:0.1488, r2:0.715, val_loss:15.0204, val_r2:-13.79
 [Pretrain] e158, train_loss:0.1473, r2:0.718, val_loss:15.1574, val_r2:-13.93
 [Pretrain] e159, train_loss:0.1471, r2:0.718, val_loss:15.5308, val_r2:-14.30
 [Pretrain] e160, train_loss:0.1470, r2:0.718, val_loss:15.6857, val_r2:-14.45
 [Pretrain] e161, train_loss:0.1470, r2:0.718, val_loss:15.9236, val_r2:-14.68

[Pretrain] e162, train_loss:0.1464, r2:0.720, val_loss:16.0940, val_r2:-14.85
 [Pretrain] e163, train_loss:0.1460, r2:0.720, val_loss:16.4572, val_r2:-15.21
 [Pretrain] e164, train_loss:0.1461, r2:0.720, val_loss:16.7381, val_r2:-15.48
 [Pretrain] e165, train_loss:0.1466, r2:0.719, val_loss:17.1158, val_r2:-15.86
 [Pretrain] e166, train_loss:0.1463, r2:0.720, val_loss:17.1405, val_r2:-15.88
 [Pretrain] e167, train_loss:0.1470, r2:0.718, val_loss:17.3950, val_r2:-16.13
 [Pretrain] e168, train_loss:0.1485, r2:0.716, val_loss:17.5573, val_r2:-16.29
 [Pretrain] e169, train_loss:0.1486, r2:0.715, val_loss:17.9153, val_r2:-16.64
 [Pretrain] e170, train_loss:0.1463, r2:0.720, val_loss:17.8806, val_r2:-16.61
 [Pretrain] e171, train_loss:0.1453, r2:0.722, val_loss:18.1132, val_r2:-16.84
 [Pretrain] e172, train_loss:0.1473, r2:0.718, val_loss:18.3738, val_r2:-17.10
 [Pretrain] e173, train_loss:0.1474, r2:0.718, val_loss:18.6022, val_r2:-17.32
 [Pretrain] e174, train_loss:0.1486, r2:0.716, val_loss:18.6323, val_r2:-17.35
 [Pretrain] e175, train_loss:0.1466, r2:0.719, val_loss:18.7666, val_r2:-17.48
 [Pretrain] e176, train_loss:0.1472, r2:0.718, val_loss:18.9949, val_r2:-17.71
 [Pretrain] e177, train_loss:0.1449, r2:0.723, val_loss:19.1651, val_r2:-17.88
 [Pretrain] e178, train_loss:0.1437, r2:0.725, val_loss:19.4683, val_r2:-18.17
 [Pretrain] e179, train_loss:0.1459, r2:0.721, val_loss:19.7519, val_r2:-18.45
 [Pretrain] e180, train_loss:0.1456, r2:0.721, val_loss:19.7571, val_r2:-18.46
 [Pretrain] e181, train_loss:0.1446, r2:0.723, val_loss:20.0268, val_r2:-18.72
 [Pretrain] e182, train_loss:0.1428, r2:0.726, val_loss:20.3989, val_r2:-19.09
 [Pretrain] e183, train_loss:0.1430, r2:0.726, val_loss:20.6104, val_r2:-19.30
 [Pretrain] e184, train_loss:0.1430, r2:0.726, val_loss:20.9684, val_r2:-19.65
 [Pretrain] e185, train_loss:0.1424, r2:0.727, val_loss:21.1290, val_r2:-19.81
 [Pretrain] e186, train_loss:0.1422, r2:0.728, val_loss:21.5923, val_r2:-20.27
 [Pretrain] e187, train_loss:0.1425, r2:0.727, val_loss:21.7799, val_r2:-20.45
 [Pretrain] e188, train_loss:0.1423, r2:0.727, val_loss:22.1516, val_r2:-20.82
 [Pretrain] e189, train_loss:0.1418, r2:0.729, val_loss:22.4833, val_r2:-21.14
 [Pretrain] e190, train_loss:0.1425, r2:0.727, val_loss:22.6958, val_r2:-21.35
 [Pretrain] e191, train_loss:0.1428, r2:0.726, val_loss:23.1389, val_r2:-21.79
 [Pretrain] e192, train_loss:0.1420, r2:0.728, val_loss:23.2834, val_r2:-21.93
 [Pretrain] e193, train_loss:0.1418, r2:0.729, val_loss:23.5683, val_r2:-22.21
 [Pretrain] e194, train_loss:0.1425, r2:0.727, val_loss:23.8182, val_r2:-22.46
 [Pretrain] e195, train_loss:0.1406, r2:0.731, val_loss:24.2874, val_r2:-22.92
 [Pretrain] e196, train_loss:0.1412, r2:0.729, val_loss:24.3026, val_r2:-22.94
 [Pretrain] e197, train_loss:0.1410, r2:0.730, val_loss:24.8566, val_r2:-23.48
 [Pretrain] e198, train_loss:0.1429, r2:0.726, val_loss:24.9799, val_r2:-23.60
 [Pretrain] e199, train_loss:0.1428, r2:0.726, val_loss:25.4014, val_r2:-24.02
 [Pretrain] e200, train_loss:0.1437, r2:0.725, val_loss:25.4618, val_r2:-24.08

[Pretrain] e201, train_loss:0.1467, r2:0.719, val_loss:25.1849, val_r2:-23.81
 [Pretrain] e202, train_loss:0.1440, r2:0.724, val_loss:25.7579, val_r2:-24.37
 [Pretrain] e203, train_loss:0.1490, r2:0.715, val_loss:25.4745, val_r2:-24.09
 [Pretrain] e204, train_loss:0.1456, r2:0.721, val_loss:25.9315, val_r2:-24.54
 [Pretrain] e205, train_loss:0.1444, r2:0.724, val_loss:25.5772, val_r2:-24.19
 [Pretrain] e206, train_loss:0.1435, r2:0.725, val_loss:25.7859, val_r2:-24.40
 [Pretrain] e207, train_loss:0.1422, r2:0.728, val_loss:26.1410, val_r2:-24.75
 [Pretrain] e208, train_loss:0.1399, r2:0.732, val_loss:26.2411, val_r2:-24.85
 [Pretrain] e209, train_loss:0.1395, r2:0.733, val_loss:26.5268, val_r2:-25.13
 [Pretrain] e210, train_loss:0.1399, r2:0.732, val_loss:26.8921, val_r2:-25.49
 [Pretrain] e211, train_loss:0.1390, r2:0.734, val_loss:27.0576, val_r2:-25.65
 [Pretrain] e212, train_loss:0.1389, r2:0.734, val_loss:27.4972, val_r2:-26.08
 [Pretrain] e213, train_loss:0.1394, r2:0.733, val_loss:27.6281, val_r2:-26.21
 [Pretrain] e214, train_loss:0.1394, r2:0.733, val_loss:27.8960, val_r2:-26.48
 [Pretrain] e215, train_loss:0.1387, r2:0.734, val_loss:28.3240, val_r2:-26.90
 [Pretrain] e216, train_loss:0.1387, r2:0.734, val_loss:28.4530, val_r2:-27.03
 [Pretrain] e217, train_loss:0.1393, r2:0.733, val_loss:28.9145, val_r2:-27.48
 [Pretrain] e218, train_loss:0.1390, r2:0.734, val_loss:28.8866, val_r2:-27.45
 [Pretrain] e219, train_loss:0.1378, r2:0.736, val_loss:29.4968, val_r2:-28.05
 [Pretrain] e220, train_loss:0.1382, r2:0.735, val_loss:29.8880, val_r2:-28.44
 [Pretrain] e221, train_loss:0.1378, r2:0.736, val_loss:29.9249, val_r2:-28.48
 [Pretrain] e222, train_loss:0.1375, r2:0.737, val_loss:30.4622, val_r2:-29.00
 [Pretrain] e223, train_loss:0.1374, r2:0.737, val_loss:30.5432, val_r2:-29.08
 [Pretrain] e224, train_loss:0.1370, r2:0.738, val_loss:30.8788, val_r2:-29.42
 [Pretrain] e225, train_loss:0.1372, r2:0.737, val_loss:31.6350, val_r2:-30.16
 [Pretrain] e226, train_loss:0.1369, r2:0.738, val_loss:31.4133, val_r2:-29.94
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 [Pretrain] e228, train_loss:0.1372, r2:0.737, val_loss:32.4740, val_r2:-30.99
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 [Pretrain] e231, train_loss:0.1370, r2:0.738, val_loss:33.3458, val_r2:-31.85
 [Pretrain] e232, train_loss:0.1369, r2:0.738, val_loss:33.5975, val_r2:-32.09
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 [Pretrain] e234, train_loss:0.1364, r2:0.739, val_loss:34.4249, val_r2:-32.91
 [Pretrain] e235, train_loss:0.1360, r2:0.740, val_loss:34.4977, val_r2:-32.98
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 [Pretrain] e237, train_loss:0.1359, r2:0.740, val_loss:35.3336, val_r2:-33.80
 [Pretrain] e238, train_loss:0.1348, r2:0.742, val_loss:35.4292, val_r2:-33.90
 [Pretrain] e239, train_loss:0.1350, r2:0.741, val_loss:36.1721, val_r2:-34.63

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 [Pretrain] e278, train_loss:0.1336, r2:0.744, val_loss:47.4308, val_r2:-45.72

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 [Pretrain] e314, train_loss:0.1294, r2:0.752, val_loss:53.9140, val_r2:-52.11
 [Pretrain] e315, train_loss:0.1304, r2:0.750, val_loss:54.1928, val_r2:-52.38
 [Pretrain] e316, train_loss:0.1299, r2:0.751, val_loss:54.7786, val_r2:-52.96
 [Pretrain] e317, train_loss:0.1304, r2:0.750, val_loss:54.5007, val_r2:-52.69

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 [Pretrain] e337, train_loss:0.1295, r2:0.752, val_loss:57.3689, val_r2:-55.51
 [Pretrain] e338, train_loss:0.1287, r2:0.754, val_loss:57.6730, val_r2:-55.81
 [Pretrain] e339, train_loss:0.1283, r2:0.754, val_loss:57.8431, val_r2:-55.98
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 [Pretrain] e342, train_loss:0.1301, r2:0.751, val_loss:57.8759, val_r2:-56.01
 [Pretrain] e343, train_loss:0.1294, r2:0.752, val_loss:58.5240, val_r2:-56.65
 [Pretrain] e344, train_loss:0.1293, r2:0.752, val_loss:58.0398, val_r2:-56.17
 [Pretrain] e345, train_loss:0.1281, r2:0.755, val_loss:58.7582, val_r2:-56.88
 [Pretrain] e346, train_loss:0.1283, r2:0.754, val_loss:58.6426, val_r2:-56.77
 [Pretrain] e347, train_loss:0.1284, r2:0.754, val_loss:58.9243, val_r2:-57.04
 [Pretrain] e348, train_loss:0.1277, r2:0.755, val_loss:58.7999, val_r2:-56.92
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 [Pretrain] e350, train_loss:0.1284, r2:0.754, val_loss:59.4446, val_r2:-57.56
 [Pretrain] e351, train_loss:0.1284, r2:0.754, val_loss:59.5251, val_r2:-57.64
 [Pretrain] e352, train_loss:0.1296, r2:0.752, val_loss:59.5086, val_r2:-57.62
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 [Pretrain] e354, train_loss:0.1282, r2:0.754, val_loss:59.5299, val_r2:-57.64
 [Pretrain] e355, train_loss:0.1281, r2:0.755, val_loss:60.0324, val_r2:-58.14
 [Pretrain] e356, train_loss:0.1293, r2:0.752, val_loss:59.8612, val_r2:-57.97

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 [Pretrain] e360, train_loss:0.1301, r2:0.751, val_loss:60.2428, val_r2:-58.34
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 [Pretrain] e387, train_loss:0.1268, r2:0.757, val_loss:63.0851, val_r2:-61.14
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 [Pretrain] e394, train_loss:0.1264, r2:0.758, val_loss:63.7725, val_r2:-61.82
 [Pretrain] e395, train_loss:0.1268, r2:0.757, val_loss:64.1166, val_r2:-62.16

[Pretrain] e396, train_loss:0.1266, r2:0.758, val_loss:64.2140, val_r2:-62.25
 [Pretrain] e397, train_loss:0.1271, r2:0.757, val_loss:64.3444, val_r2:-62.38
 [Pretrain] e398, train_loss:0.1279, r2:0.755, val_loss:64.2420, val_r2:-62.28
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 [Pretrain] e404, train_loss:0.1278, r2:0.755, val_loss:65.5310, val_r2:-63.55
 [Pretrain] e405, train_loss:0.1269, r2:0.757, val_loss:65.4209, val_r2:-63.44
 [Pretrain] e406, train_loss:0.1275, r2:0.756, val_loss:65.7691, val_r2:-63.79
 [Pretrain] e407, train_loss:0.1295, r2:0.752, val_loss:65.4543, val_r2:-63.48
 [Pretrain] e408, train_loss:0.1289, r2:0.753, val_loss:65.2492, val_r2:-63.27
 [Pretrain] e409, train_loss:0.1299, r2:0.751, val_loss:65.3655, val_r2:-63.39
 [Pretrain] e410, train_loss:0.1293, r2:0.752, val_loss:64.9982, val_r2:-63.03
 [Pretrain] e411, train_loss:0.1299, r2:0.751, val_loss:65.4137, val_r2:-63.44
 [Pretrain] e412, train_loss:0.1314, r2:0.748, val_loss:64.7730, val_r2:-62.81
 [Pretrain] e413, train_loss:0.1317, r2:0.748, val_loss:65.0155, val_r2:-63.04
 [Pretrain] e414, train_loss:0.1308, r2:0.750, val_loss:64.6001, val_r2:-62.63
 [Pretrain] e415, train_loss:0.1320, r2:0.747, val_loss:64.2739, val_r2:-62.31
 [Pretrain] e416, train_loss:0.1283, r2:0.754, val_loss:64.3614, val_r2:-62.40
 [Pretrain] e417, train_loss:0.1286, r2:0.754, val_loss:64.2459, val_r2:-62.29
 [Pretrain] e418, train_loss:0.1286, r2:0.754, val_loss:64.3561, val_r2:-62.39
 [Pretrain] e419, train_loss:0.1271, r2:0.757, val_loss:64.7579, val_r2:-62.79
 [Pretrain] e420, train_loss:0.1263, r2:0.758, val_loss:64.7430, val_r2:-62.78
 [Pretrain] e421, train_loss:0.1267, r2:0.757, val_loss:64.6968, val_r2:-62.73
 [Pretrain] e422, train_loss:0.1260, r2:0.759, val_loss:65.0069, val_r2:-63.04
 [Pretrain] e423, train_loss:0.1266, r2:0.758, val_loss:64.9563, val_r2:-62.99
 [Pretrain] e424, train_loss:0.1264, r2:0.758, val_loss:65.3149, val_r2:-63.34
 [Pretrain] e425, train_loss:0.1263, r2:0.758, val_loss:66.0127, val_r2:-64.03
 [Pretrain] e426, train_loss:0.1286, r2:0.754, val_loss:66.0286, val_r2:-64.04
 [Pretrain] e427, train_loss:0.1282, r2:0.755, val_loss:65.6442, val_r2:-63.66
 [Pretrain] e428, train_loss:0.1287, r2:0.754, val_loss:65.2718, val_r2:-63.30
 [Pretrain] e429, train_loss:0.1284, r2:0.754, val_loss:66.0917, val_r2:-64.10
 [Pretrain] e430, train_loss:0.1286, r2:0.754, val_loss:65.9789, val_r2:-63.99
 [Pretrain] e431, train_loss:0.1270, r2:0.757, val_loss:65.6332, val_r2:-63.65
 [Pretrain] e432, train_loss:0.1275, r2:0.756, val_loss:65.9951, val_r2:-64.01
 [Pretrain] e433, train_loss:0.1278, r2:0.755, val_loss:65.7779, val_r2:-63.80
 [Pretrain] e434, train_loss:0.1270, r2:0.757, val_loss:65.8656, val_r2:-63.88

[Pretrain] e435, train_loss:0.1269, r2:0.757, val_loss:66.2140, val_r2:-64.22
 [Pretrain] e436, train_loss:0.1272, r2:0.756, val_loss:65.8386, val_r2:-63.85
 [Pretrain] e437, train_loss:0.1262, r2:0.758, val_loss:66.1535, val_r2:-64.17
 [Pretrain] e438, train_loss:0.1266, r2:0.758, val_loss:66.3945, val_r2:-64.40
 [Pretrain] e439, train_loss:0.1266, r2:0.757, val_loss:66.5817, val_r2:-64.59
 [Pretrain] e440, train_loss:0.1272, r2:0.756, val_loss:66.0829, val_r2:-64.10
 [Pretrain] e441, train_loss:0.1273, r2:0.756, val_loss:66.6331, val_r2:-64.64
 [Pretrain] e442, train_loss:0.1263, r2:0.758, val_loss:66.5014, val_r2:-64.51
 [Pretrain] e443, train_loss:0.1262, r2:0.758, val_loss:66.4983, val_r2:-64.50
 [Pretrain] e444, train_loss:0.1259, r2:0.759, val_loss:66.6279, val_r2:-64.63
 [Pretrain] e445, train_loss:0.1258, r2:0.759, val_loss:66.8192, val_r2:-64.82
 [Pretrain] e446, train_loss:0.1264, r2:0.758, val_loss:67.1713, val_r2:-65.17
 [Pretrain] e447, train_loss:0.1256, r2:0.759, val_loss:66.9738, val_r2:-64.97
 [Pretrain] e448, train_loss:0.1260, r2:0.759, val_loss:66.8433, val_r2:-64.84
 [Pretrain] e449, train_loss:0.1257, r2:0.759, val_loss:67.5193, val_r2:-65.51
 [Pretrain] e450, train_loss:0.1269, r2:0.757, val_loss:66.9600, val_r2:-64.96
 [Pretrain] e451, train_loss:0.1265, r2:0.758, val_loss:67.3108, val_r2:-65.31
 [Pretrain] e452, train_loss:0.1251, r2:0.760, val_loss:67.3677, val_r2:-65.36
 [Pretrain] e453, train_loss:0.1251, r2:0.760, val_loss:67.2560, val_r2:-65.25
 [Pretrain] e454, train_loss:0.1254, r2:0.760, val_loss:67.7457, val_r2:-65.73
 [Pretrain] e455, train_loss:0.1264, r2:0.758, val_loss:67.9117, val_r2:-65.90
 [Pretrain] e456, train_loss:0.1271, r2:0.757, val_loss:67.7385, val_r2:-65.73
 [Pretrain] e457, train_loss:0.1265, r2:0.758, val_loss:67.9261, val_r2:-65.91
 [Pretrain] e458, train_loss:0.1275, r2:0.756, val_loss:67.5707, val_r2:-65.56
 [Pretrain] e459, train_loss:0.1273, r2:0.756, val_loss:67.7746, val_r2:-65.76
 [Pretrain] e460, train_loss:0.1268, r2:0.757, val_loss:68.3272, val_r2:-66.31
 [Pretrain] e461, train_loss:0.1271, r2:0.757, val_loss:67.5443, val_r2:-65.54
 [Pretrain] e462, train_loss:0.1260, r2:0.759, val_loss:68.0481, val_r2:-66.03
 [Pretrain] e463, train_loss:0.1254, r2:0.760, val_loss:68.2228, val_r2:-66.20
 [Pretrain] e464, train_loss:0.1250, r2:0.761, val_loss:68.1244, val_r2:-66.11
 [Pretrain] e465, train_loss:0.1259, r2:0.759, val_loss:68.2036, val_r2:-66.18
 [Pretrain] e466, train_loss:0.1261, r2:0.758, val_loss:68.3883, val_r2:-66.37
 [Pretrain] e467, train_loss:0.1266, r2:0.758, val_loss:68.2160, val_r2:-66.20
 [Pretrain] e468, train_loss:0.1251, r2:0.760, val_loss:68.3553, val_r2:-66.33
 [Pretrain] e469, train_loss:0.1252, r2:0.760, val_loss:68.4945, val_r2:-66.47
 [Pretrain] e470, train_loss:0.1252, r2:0.760, val_loss:68.7112, val_r2:-66.68
 [Pretrain] e471, train_loss:0.1253, r2:0.760, val_loss:68.6595, val_r2:-66.63
 [Pretrain] e472, train_loss:0.1250, r2:0.761, val_loss:68.7021, val_r2:-66.68
 [Pretrain] e473, train_loss:0.1259, r2:0.759, val_loss:68.8211, val_r2:-66.79

[Pretrain] e474, train_loss:0.1258, r2:0.759, val_loss:69.0155, val_r2:-66.98
 [Pretrain] e475, train_loss:0.1259, r2:0.759, val_loss:69.1999, val_r2:-67.17
 [Pretrain] e476, train_loss:0.1269, r2:0.757, val_loss:69.0265, val_r2:-67.00
 [Pretrain] e477, train_loss:0.1255, r2:0.760, val_loss:68.9221, val_r2:-66.89
 [Pretrain] e478, train_loss:0.1257, r2:0.759, val_loss:69.4046, val_r2:-67.37
 [Pretrain] e479, train_loss:0.1261, r2:0.758, val_loss:68.9786, val_r2:-66.95
 [Pretrain] e480, train_loss:0.1252, r2:0.760, val_loss:69.1451, val_r2:-67.11
 [Pretrain] e481, train_loss:0.1256, r2:0.759, val_loss:69.2499, val_r2:-67.22
 [Pretrain] e482, train_loss:0.1259, r2:0.759, val_loss:69.4793, val_r2:-67.44
 [Pretrain] e483, train_loss:0.1269, r2:0.757, val_loss:68.8234, val_r2:-66.80
 [Pretrain] e484, train_loss:0.1263, r2:0.758, val_loss:69.3500, val_r2:-67.31
 [Pretrain] e485, train_loss:0.1259, r2:0.759, val_loss:69.2952, val_r2:-67.26
 [Pretrain] e486, train_loss:0.1254, r2:0.760, val_loss:69.0522, val_r2:-67.02
 [Pretrain] e487, train_loss:0.1250, r2:0.761, val_loss:69.5638, val_r2:-67.52
 [Pretrain] e488, train_loss:0.1245, r2:0.762, val_loss:69.8155, val_r2:-67.77
 [Pretrain] e489, train_loss:0.1255, r2:0.760, val_loss:69.6154, val_r2:-67.58
 [Pretrain] e490, train_loss:0.1253, r2:0.760, val_loss:69.4240, val_r2:-67.39
 [Pretrain] e491, train_loss:0.1257, r2:0.759, val_loss:69.9029, val_r2:-67.86
 [Pretrain] e492, train_loss:0.1250, r2:0.761, val_loss:69.4456, val_r2:-67.41
 [Pretrain] e493, train_loss:0.1264, r2:0.758, val_loss:70.2242, val_r2:-68.18
 [Pretrain] e494, train_loss:0.1285, r2:0.754, val_loss:69.7321, val_r2:-67.69
 [Pretrain] e495, train_loss:0.1274, r2:0.756, val_loss:69.4337, val_r2:-67.40
 [Pretrain] e496, train_loss:0.1269, r2:0.757, val_loss:69.3830, val_r2:-67.35
 [Pretrain] e497, train_loss:0.1257, r2:0.759, val_loss:69.6036, val_r2:-67.56
 [Pretrain] e498, train_loss:0.1252, r2:0.760, val_loss:69.6049, val_r2:-67.57
 [Pretrain] e499, train_loss:0.1246, r2:0.761, val_loss:69.4308, val_r2:-67.39
 [Finetune] e0, train_loss:69.9311, r2:-68.019, val_loss:55.3586, val_r2:-53.5
 [Finetune] e1, train_loss:55.5204, r2:-53.796, val_loss:39.9479, val_r2:-38.3
 [Finetune] e2, train_loss:39.9380, r2:-38.417, val_loss:26.9371, val_r2:-25.5
 [Finetune] e3, train_loss:26.7799, r2:-25.431, val_loss:17.5354, val_r2:-16.2
 [Finetune] e4, train_loss:17.2834, r2:-16.058, val_loss:11.4745, val_r2:-10.3
 [Finetune] e5, train_loss:11.3063, r2:-10.159, val_loss:7.5697, val_r2:-6.457
 [Finetune] e6, train_loss:7.8231, r2:-6.721, val_loss:5.2097, val_r2:-4.132
 [Finetune] e7, train_loss:5.4792, r2:-4.408, val_loss:3.8959, val_r2:-2.838
 [Finetune] e8, train_loss:3.8888, r2:-2.838, val_loss:3.2539, val_r2:-2.205
 [Finetune] e9, train_loss:2.8924, r2:-1.855, val_loss:3.0144, val_r2:-1.970
 [Finetune] e10, train_loss:2.3260, r2:-1.296, val_loss:2.9666, val_r2:-1.922
 [Finetune] e11, train_loss:2.0893, r2:-1.062, val_loss:2.9721, val_r2:-1.928
 [Finetune] e12, train_loss:2.0346, r2:-1.008, val_loss:2.9321, val_r2:-1.889

[Finetune] e13, train_loss:2.0210, r2:-0.995, val_loss:2.8063, val_r2:-1.765
 [Finetune] e14, train_loss:1.9830, r2:-0.957, val_loss:2.5789, val_r2:-1.540
 [Finetune] e15, train_loss:1.8643, r2:-0.840, val_loss:2.2606, val_r2:-1.227
 [Finetune] e16, train_loss:1.6498, r2:-0.628, val_loss:1.8907, val_r2:-0.863
 [Finetune] e17, train_loss:1.3678, r2:-0.350, val_loss:1.4936, val_r2:-0.471
 [Finetune] e18, train_loss:1.0560, r2:-0.042, val_loss:1.1110, val_r2:-0.095
 [Finetune] e19, train_loss:0.7640, r2:0.246, val_loss:0.7897, val_r2:0.222
 [Finetune] e20, train_loss:0.5287, r2:0.478, val_loss:0.5561, val_r2:0.452
 Model successfully saved
 [Finetune] e21, train_loss:0.3676, r2:0.637, val_loss:0.4126, val_r2:0.594
 Model successfully saved
 [Finetune] e22, train_loss:0.2768, r2:0.727, val_loss:0.3458, val_r2:0.659
 Model successfully saved
 [Finetune] e23, train_loss:0.2401, r2:0.763, val_loss:0.3384, val_r2:0.667
 Model successfully saved
 [Finetune] e24, train_loss:0.2439, r2:0.759, val_loss:0.3682, val_r2:0.637
 [Finetune] e25, train_loss:0.2668, r2:0.737, val_loss:0.4106, val_r2:0.596
 [Finetune] e26, train_loss:0.2898, r2:0.714, val_loss:0.4469, val_r2:0.560
 [Finetune] e27, train_loss:0.3024, r2:0.702, val_loss:0.4693, val_r2:0.538
 [Finetune] e28, train_loss:0.3027, r2:0.701, val_loss:0.4761, val_r2:0.531
 [Finetune] e29, train_loss:0.2947, r2:0.709, val_loss:0.4706, val_r2:0.536
 [Finetune] e30, train_loss:0.2823, r2:0.721, val_loss:0.4568, val_r2:0.550
 [Finetune] e31, train_loss:0.2687, r2:0.735, val_loss:0.4385, val_r2:0.568
 [Finetune] e32, train_loss:0.2552, r2:0.748, val_loss:0.4185, val_r2:0.588
 [Finetune] e33, train_loss:0.2415, r2:0.762, val_loss:0.3978, val_r2:0.608
 [Finetune] e34, train_loss:0.2261, r2:0.777, val_loss:0.3767, val_r2:0.629
 [Finetune] e35, train_loss:0.2086, r2:0.794, val_loss:0.3555, val_r2:0.650
 [Finetune] e36, train_loss:0.1884, r2:0.814, val_loss:0.3346, val_r2:0.670
 Model successfully saved
 [Finetune] e37, train_loss:0.1665, r2:0.836, val_loss:0.3148, val_r2:0.690
 Model successfully saved
 [Finetune] e38, train_loss:0.1443, r2:0.858, val_loss:0.2973, val_r2:0.707
 Model successfully saved
 [Finetune] e39, train_loss:0.1238, r2:0.878, val_loss:0.2827, val_r2:0.721
 Model successfully saved
 [Finetune] e40, train_loss:0.1065, r2:0.895, val_loss:0.2719, val_r2:0.732
 Model successfully saved
 [Finetune] e41, train_loss:0.0937, r2:0.908, val_loss:0.2647, val_r2:0.739
 Model successfully saved

[Finetune] e42, train_loss:0.0860, r2:0.915, val_loss:0.2608, val_r2:0.743
Model successfully saved

[Finetune] e43, train_loss:0.0828, r2:0.918, val_loss:0.2595, val_r2:0.744
Model successfully saved

[Finetune] e44, train_loss:0.0828, r2:0.918, val_loss:0.2597, val_r2:0.744

[Finetune] e45, train_loss:0.0842, r2:0.917, val_loss:0.2608, val_r2:0.743

[Finetune] e46, train_loss:0.0858, r2:0.915, val_loss:0.2622, val_r2:0.742

[Finetune] e47, train_loss:0.0865, r2:0.915, val_loss:0.2634, val_r2:0.741

[Finetune] e48, train_loss:0.0859, r2:0.915, val_loss:0.2644, val_r2:0.740

[Finetune] e49, train_loss:0.0840, r2:0.917, val_loss:0.2652, val_r2:0.739

[Finetune] e50, train_loss:0.0811, r2:0.920, val_loss:0.2659, val_r2:0.738

[Finetune] e51, train_loss:0.0777, r2:0.923, val_loss:0.2667, val_r2:0.737

[Finetune] e52, train_loss:0.0744, r2:0.927, val_loss:0.2675, val_r2:0.737

[Finetune] e53, train_loss:0.0715, r2:0.929, val_loss:0.2682, val_r2:0.736

[Finetune] e54, train_loss:0.0690, r2:0.932, val_loss:0.2685, val_r2:0.735

[Finetune] e55, train_loss:0.0670, r2:0.934, val_loss:0.2683, val_r2:0.736

[Finetune] e56, train_loss:0.0651, r2:0.936, val_loss:0.2674, val_r2:0.737

[Finetune] e57, train_loss:0.0633, r2:0.938, val_loss:0.2657, val_r2:0.738

[Finetune] e58, train_loss:0.0613, r2:0.940, val_loss:0.2633, val_r2:0.741

[Finetune] e59, train_loss:0.0591, r2:0.942, val_loss:0.2604, val_r2:0.743

[Finetune] e60, train_loss:0.0566, r2:0.944, val_loss:0.2574, val_r2:0.746
Model successfully saved

[Finetune] e61, train_loss:0.0542, r2:0.946, val_loss:0.2544, val_r2:0.749
Model successfully saved

[Finetune] e62, train_loss:0.0520, r2:0.949, val_loss:0.2518, val_r2:0.752
Model successfully saved

[Finetune] e63, train_loss:0.0501, r2:0.951, val_loss:0.2496, val_r2:0.754
Model successfully saved

[Finetune] e64, train_loss:0.0487, r2:0.952, val_loss:0.2480, val_r2:0.756
Model successfully saved

[Finetune] e65, train_loss:0.0476, r2:0.953, val_loss:0.2469, val_r2:0.757
Model successfully saved

[Finetune] e66, train_loss:0.0469, r2:0.954, val_loss:0.2462, val_r2:0.757
Model successfully saved

[Finetune] e67, train_loss:0.0464, r2:0.954, val_loss:0.2457, val_r2:0.758
Model successfully saved

[Finetune] e68, train_loss:0.0460, r2:0.955, val_loss:0.2453, val_r2:0.758
Model successfully saved

[Finetune] e69, train_loss:0.0454, r2:0.955, val_loss:0.2450, val_r2:0.759

Model successfully saved

[Finetune] e70, train_loss:0.0447, r2:0.956, val_loss:0.2448, val_r2:0.759

Model successfully saved

[Finetune] e71, train_loss:0.0438, r2:0.957, val_loss:0.2446, val_r2:0.759

Model successfully saved

[Finetune] e72, train_loss:0.0429, r2:0.958, val_loss:0.2445, val_r2:0.759

Model successfully saved

[Finetune] e73, train_loss:0.0419, r2:0.959, val_loss:0.2446, val_r2:0.759

[Finetune] e74, train_loss:0.0410, r2:0.960, val_loss:0.2448, val_r2:0.759

[Finetune] e75, train_loss:0.0401, r2:0.960, val_loss:0.2451, val_r2:0.759

[Finetune] e76, train_loss:0.0393, r2:0.961, val_loss:0.2456, val_r2:0.758

[Finetune] e77, train_loss:0.0387, r2:0.962, val_loss:0.2461, val_r2:0.758

[Finetune] e78, train_loss:0.0380, r2:0.962, val_loss:0.2467, val_r2:0.757

[Finetune] e79, train_loss:0.0374, r2:0.963, val_loss:0.2472, val_r2:0.756

[Finetune] e80, train_loss:0.0368, r2:0.964, val_loss:0.2478, val_r2:0.756

[Finetune] e81, train_loss:0.0362, r2:0.964, val_loss:0.2483, val_r2:0.755

[Finetune] e82, train_loss:0.0356, r2:0.965, val_loss:0.2488, val_r2:0.755

[Finetune] e83, train_loss:0.0350, r2:0.965, val_loss:0.2494, val_r2:0.754

[Finetune] e84, train_loss:0.0345, r2:0.966, val_loss:0.2500, val_r2:0.754

[Finetune] e85, train_loss:0.0340, r2:0.966, val_loss:0.2505, val_r2:0.753

[Finetune] e86, train_loss:0.0336, r2:0.967, val_loss:0.2511, val_r2:0.753

[Finetune] e87, train_loss:0.0332, r2:0.967, val_loss:0.2517, val_r2:0.752

[Finetune] e88, train_loss:0.0328, r2:0.968, val_loss:0.2523, val_r2:0.751

[Finetune] e89, train_loss:0.0325, r2:0.968, val_loss:0.2529, val_r2:0.751

[Finetune] e90, train_loss:0.0321, r2:0.968, val_loss:0.2534, val_r2:0.750

[Finetune] e91, train_loss:0.0317, r2:0.969, val_loss:0.2539, val_r2:0.750

[Finetune] e92, train_loss:0.0314, r2:0.969, val_loss:0.2543, val_r2:0.749

[Finetune] e93, train_loss:0.0310, r2:0.969, val_loss:0.2547, val_r2:0.749

[Finetune] e94, train_loss:0.0306, r2:0.970, val_loss:0.2550, val_r2:0.749

[Finetune] e95, train_loss:0.0302, r2:0.970, val_loss:0.2553, val_r2:0.748

[Finetune] e96, train_loss:0.0299, r2:0.971, val_loss:0.2556, val_r2:0.748

[Finetune] e97, train_loss:0.0296, r2:0.971, val_loss:0.2558, val_r2:0.748

[Finetune] e98, train_loss:0.0292, r2:0.971, val_loss:0.2560, val_r2:0.748

[Finetune] e99, train_loss:0.0289, r2:0.971, val_loss:0.2562, val_r2:0.748

[Finetune] e100, train_loss:0.0286, r2:0.972, val_loss:0.2563, val_r2:0.748

[Finetune] e101, train_loss:0.0283, r2:0.972, val_loss:0.2564, val_r2:0.747

[Finetune] e102, train_loss:0.0280, r2:0.972, val_loss:0.2565, val_r2:0.747

[Finetune] e103, train_loss:0.0277, r2:0.973, val_loss:0.2565, val_r2:0.747

[Finetune] e104, train_loss:0.0274, r2:0.973, val_loss:0.2566, val_r2:0.747

[Finetune] e105, train_loss:0.0272, r2:0.973, val_loss:0.2567, val_r2:0.747
[Finetune] e106, train_loss:0.0269, r2:0.973, val_loss:0.2569, val_r2:0.747
[Finetune] e107, train_loss:0.0267, r2:0.974, val_loss:0.2570, val_r2:0.747
[Finetune] e108, train_loss:0.0264, r2:0.974, val_loss:0.2571, val_r2:0.747
[Finetune] e109, train_loss:0.0262, r2:0.974, val_loss:0.2573, val_r2:0.747
[Finetune] e110, train_loss:0.0260, r2:0.974, val_loss:0.2575, val_r2:0.746
[Finetune] e111, train_loss:0.0257, r2:0.975, val_loss:0.2577, val_r2:0.746
[Finetune] e112, train_loss:0.0255, r2:0.975, val_loss:0.2579, val_r2:0.746
[Finetune] e113, train_loss:0.0253, r2:0.975, val_loss:0.2582, val_r2:0.746
[Finetune] e114, train_loss:0.0250, r2:0.975, val_loss:0.2585, val_r2:0.745
[Finetune] e115, train_loss:0.0248, r2:0.975, val_loss:0.2587, val_r2:0.745
[Finetune] e116, train_loss:0.0246, r2:0.976, val_loss:0.2590, val_r2:0.745
[Finetune] e117, train_loss:0.0244, r2:0.976, val_loss:0.2593, val_r2:0.745
[Finetune] e118, train_loss:0.0242, r2:0.976, val_loss:0.2596, val_r2:0.744
[Finetune] e119, train_loss:0.0240, r2:0.976, val_loss:0.2599, val_r2:0.744
[Finetune] e120, train_loss:0.0238, r2:0.977, val_loss:0.2602, val_r2:0.744
[Finetune] e121, train_loss:0.0236, r2:0.977, val_loss:0.2605, val_r2:0.743
[Finetune] e122, train_loss:0.0234, r2:0.977, val_loss:0.2608, val_r2:0.743
[Finetune] e123, train_loss:0.0232, r2:0.977, val_loss:0.2611, val_r2:0.743
[Finetune] e124, train_loss:0.0231, r2:0.977, val_loss:0.2613, val_r2:0.743
[Finetune] e125, train_loss:0.0229, r2:0.977, val_loss:0.2616, val_r2:0.742
[Finetune] e126, train_loss:0.0227, r2:0.978, val_loss:0.2618, val_r2:0.742
[Finetune] e127, train_loss:0.0225, r2:0.978, val_loss:0.2620, val_r2:0.742
[Finetune] e128, train_loss:0.0224, r2:0.978, val_loss:0.2623, val_r2:0.742
[Finetune] e129, train_loss:0.0222, r2:0.978, val_loss:0.2625, val_r2:0.741
[Finetune] e130, train_loss:0.0220, r2:0.978, val_loss:0.2626, val_r2:0.741
[Finetune] e131, train_loss:0.0219, r2:0.978, val_loss:0.2628, val_r2:0.741
[Finetune] e132, train_loss:0.0217, r2:0.979, val_loss:0.2630, val_r2:0.741
[Finetune] e133, train_loss:0.0215, r2:0.979, val_loss:0.2632, val_r2:0.741
[Finetune] e134, train_loss:0.0214, r2:0.979, val_loss:0.2633, val_r2:0.741
[Finetune] e135, train_loss:0.0212, r2:0.979, val_loss:0.2635, val_r2:0.740
[Finetune] e136, train_loss:0.0211, r2:0.979, val_loss:0.2636, val_r2:0.740
[Finetune] e137, train_loss:0.0209, r2:0.979, val_loss:0.2638, val_r2:0.740
[Finetune] e138, train_loss:0.0208, r2:0.979, val_loss:0.2640, val_r2:0.740
[Finetune] e139, train_loss:0.0206, r2:0.980, val_loss:0.2641, val_r2:0.740
[Finetune] e140, train_loss:0.0205, r2:0.980, val_loss:0.2643, val_r2:0.740
[Finetune] e141, train_loss:0.0204, r2:0.980, val_loss:0.2644, val_r2:0.740
[Finetune] e142, train_loss:0.0202, r2:0.980, val_loss:0.2646, val_r2:0.739
[Finetune] e143, train_loss:0.0201, r2:0.980, val_loss:0.2647, val_r2:0.739

[Finetune] e144, train_loss:0.0200, r2:0.980, val_loss:0.2649, val_r2:0.739
[Finetune] e145, train_loss:0.0198, r2:0.980, val_loss:0.2650, val_r2:0.739
[Finetune] e146, train_loss:0.0197, r2:0.981, val_loss:0.2652, val_r2:0.739
[Finetune] e147, train_loss:0.0196, r2:0.981, val_loss:0.2653, val_r2:0.739
[Finetune] e148, train_loss:0.0194, r2:0.981, val_loss:0.2655, val_r2:0.738
[Finetune] e149, train_loss:0.0193, r2:0.981, val_loss:0.2656, val_r2:0.738
[Finetune] e150, train_loss:0.0192, r2:0.981, val_loss:0.2658, val_r2:0.738
[Finetune] e151, train_loss:0.0191, r2:0.981, val_loss:0.2659, val_r2:0.738
[Finetune] e152, train_loss:0.0189, r2:0.981, val_loss:0.2661, val_r2:0.738
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[Finetune] e154, train_loss:0.0187, r2:0.982, val_loss:0.2664, val_r2:0.738
[Finetune] e155, train_loss:0.0186, r2:0.982, val_loss:0.2666, val_r2:0.737
[Finetune] e156, train_loss:0.0184, r2:0.982, val_loss:0.2667, val_r2:0.737
[Finetune] e157, train_loss:0.0183, r2:0.982, val_loss:0.2669, val_r2:0.737
[Finetune] e158, train_loss:0.0182, r2:0.982, val_loss:0.2671, val_r2:0.737
[Finetune] e159, train_loss:0.0181, r2:0.982, val_loss:0.2672, val_r2:0.737
[Finetune] e160, train_loss:0.0180, r2:0.982, val_loss:0.2674, val_r2:0.737
[Finetune] e161, train_loss:0.0178, r2:0.982, val_loss:0.2676, val_r2:0.736
[Finetune] e162, train_loss:0.0177, r2:0.982, val_loss:0.2677, val_r2:0.736
[Finetune] e163, train_loss:0.0176, r2:0.983, val_loss:0.2679, val_r2:0.736
[Finetune] e164, train_loss:0.0175, r2:0.983, val_loss:0.2680, val_r2:0.736
[Finetune] e165, train_loss:0.0174, r2:0.983, val_loss:0.2682, val_r2:0.736
[Finetune] e166, train_loss:0.0173, r2:0.983, val_loss:0.2683, val_r2:0.736
[Finetune] e167, train_loss:0.0172, r2:0.983, val_loss:0.2685, val_r2:0.735
[Finetune] e168, train_loss:0.0171, r2:0.983, val_loss:0.2686, val_r2:0.735
[Finetune] e169, train_loss:0.0170, r2:0.983, val_loss:0.2688, val_r2:0.735
[Finetune] e170, train_loss:0.0169, r2:0.983, val_loss:0.2689, val_r2:0.735
[Finetune] e171, train_loss:0.0168, r2:0.983, val_loss:0.2691, val_r2:0.735
[Finetune] e172, train_loss:0.0167, r2:0.984, val_loss:0.2693, val_r2:0.735
[Finetune] e173, train_loss:0.0166, r2:0.984, val_loss:0.2694, val_r2:0.735
[Finetune] e174, train_loss:0.0165, r2:0.984, val_loss:0.2696, val_r2:0.734
[Finetune] e175, train_loss:0.0164, r2:0.984, val_loss:0.2698, val_r2:0.734
[Finetune] e176, train_loss:0.0163, r2:0.984, val_loss:0.2700, val_r2:0.734
[Finetune] e177, train_loss:0.0162, r2:0.984, val_loss:0.2701, val_r2:0.734
[Finetune] e178, train_loss:0.0161, r2:0.984, val_loss:0.2703, val_r2:0.734
[Finetune] e179, train_loss:0.0160, r2:0.984, val_loss:0.2705, val_r2:0.733
[Finetune] e180, train_loss:0.0159, r2:0.984, val_loss:0.2707, val_r2:0.733
[Finetune] e181, train_loss:0.0158, r2:0.984, val_loss:0.2709, val_r2:0.733
[Finetune] e182, train_loss:0.0157, r2:0.985, val_loss:0.2711, val_r2:0.733

[Finetune] e183, train_loss:0.0156, r2:0.985, val_loss:0.2712, val_r2:0.733
[Finetune] e184, train_loss:0.0155, r2:0.985, val_loss:0.2714, val_r2:0.733
[Finetune] e185, train_loss:0.0154, r2:0.985, val_loss:0.2715, val_r2:0.732
[Finetune] e186, train_loss:0.0153, r2:0.985, val_loss:0.2717, val_r2:0.732
[Finetune] e187, train_loss:0.0153, r2:0.985, val_loss:0.2718, val_r2:0.732
[Finetune] e188, train_loss:0.0152, r2:0.985, val_loss:0.2720, val_r2:0.732
[Finetune] e189, train_loss:0.0151, r2:0.985, val_loss:0.2721, val_r2:0.732
[Finetune] e190, train_loss:0.0150, r2:0.985, val_loss:0.2723, val_r2:0.732
[Finetune] e191, train_loss:0.0149, r2:0.985, val_loss:0.2724, val_r2:0.732
[Finetune] e192, train_loss:0.0148, r2:0.985, val_loss:0.2726, val_r2:0.731
[Finetune] e193, train_loss:0.0148, r2:0.985, val_loss:0.2727, val_r2:0.731
[Finetune] e194, train_loss:0.0147, r2:0.986, val_loss:0.2729, val_r2:0.731
[Finetune] e195, train_loss:0.0146, r2:0.986, val_loss:0.2730, val_r2:0.731
[Finetune] e196, train_loss:0.0145, r2:0.986, val_loss:0.2732, val_r2:0.731
[Finetune] e197, train_loss:0.0144, r2:0.986, val_loss:0.2734, val_r2:0.731
[Finetune] e198, train_loss:0.0144, r2:0.986, val_loss:0.2735, val_r2:0.731
[Finetune] e199, train_loss:0.0143, r2:0.986, val_loss:0.2737, val_r2:0.730
[Finetune] e200, train_loss:0.0142, r2:0.986, val_loss:0.2738, val_r2:0.730
[Finetune] e201, train_loss:0.0141, r2:0.986, val_loss:0.2740, val_r2:0.730
[Finetune] e202, train_loss:0.0141, r2:0.986, val_loss:0.2741, val_r2:0.730
[Finetune] e203, train_loss:0.0140, r2:0.986, val_loss:0.2743, val_r2:0.730
[Finetune] e204, train_loss:0.0139, r2:0.986, val_loss:0.2744, val_r2:0.730
[Finetune] e205, train_loss:0.0139, r2:0.986, val_loss:0.2746, val_r2:0.730
[Finetune] e206, train_loss:0.0138, r2:0.986, val_loss:0.2747, val_r2:0.729
[Finetune] e207, train_loss:0.0137, r2:0.986, val_loss:0.2749, val_r2:0.729
[Finetune] e208, train_loss:0.0137, r2:0.987, val_loss:0.2750, val_r2:0.729
[Finetune] e209, train_loss:0.0136, r2:0.987, val_loss:0.2752, val_r2:0.729
[Finetune] e210, train_loss:0.0135, r2:0.987, val_loss:0.2753, val_r2:0.729
[Finetune] e211, train_loss:0.0135, r2:0.987, val_loss:0.2754, val_r2:0.729
[Finetune] e212, train_loss:0.0134, r2:0.987, val_loss:0.2756, val_r2:0.729
[Finetune] e213, train_loss:0.0133, r2:0.987, val_loss:0.2757, val_r2:0.728
[Finetune] e214, train_loss:0.0133, r2:0.987, val_loss:0.2758, val_r2:0.728
[Finetune] e215, train_loss:0.0132, r2:0.987, val_loss:0.2760, val_r2:0.728
[Finetune] e216, train_loss:0.0132, r2:0.987, val_loss:0.2761, val_r2:0.728
[Finetune] e217, train_loss:0.0131, r2:0.987, val_loss:0.2762, val_r2:0.728
[Finetune] e218, train_loss:0.0130, r2:0.987, val_loss:0.2764, val_r2:0.728
[Finetune] e219, train_loss:0.0130, r2:0.987, val_loss:0.2765, val_r2:0.728
[Finetune] e220, train_loss:0.0129, r2:0.987, val_loss:0.2766, val_r2:0.728
[Finetune] e221, train_loss:0.0129, r2:0.987, val_loss:0.2767, val_r2:0.727

[Finetune] e222, train_loss:0.0128, r2:0.987, val_loss:0.2769, val_r2:0.727
[Finetune] e223, train_loss:0.0128, r2:0.987, val_loss:0.2770, val_r2:0.727
[Finetune] e224, train_loss:0.0127, r2:0.987, val_loss:0.2771, val_r2:0.727
[Finetune] e225, train_loss:0.0126, r2:0.988, val_loss:0.2772, val_r2:0.727
[Finetune] e226, train_loss:0.0126, r2:0.988, val_loss:0.2773, val_r2:0.727
[Finetune] e227, train_loss:0.0125, r2:0.988, val_loss:0.2774, val_r2:0.727
[Finetune] e228, train_loss:0.0125, r2:0.988, val_loss:0.2775, val_r2:0.727
[Finetune] e229, train_loss:0.0124, r2:0.988, val_loss:0.2776, val_r2:0.727
[Finetune] e230, train_loss:0.0124, r2:0.988, val_loss:0.2777, val_r2:0.726
[Finetune] e231, train_loss:0.0123, r2:0.988, val_loss:0.2778, val_r2:0.726
[Finetune] e232, train_loss:0.0123, r2:0.988, val_loss:0.2779, val_r2:0.726
[Finetune] e233, train_loss:0.0122, r2:0.988, val_loss:0.2780, val_r2:0.726
[Finetune] e234, train_loss:0.0122, r2:0.988, val_loss:0.2781, val_r2:0.726
[Finetune] e235, train_loss:0.0121, r2:0.988, val_loss:0.2782, val_r2:0.726
[Finetune] e236, train_loss:0.0121, r2:0.988, val_loss:0.2783, val_r2:0.726
[Finetune] e237, train_loss:0.0120, r2:0.988, val_loss:0.2784, val_r2:0.726
[Finetune] e238, train_loss:0.0119, r2:0.988, val_loss:0.2785, val_r2:0.726
[Finetune] e239, train_loss:0.0119, r2:0.988, val_loss:0.2786, val_r2:0.726
[Finetune] e240, train_loss:0.0118, r2:0.988, val_loss:0.2787, val_r2:0.725
[Finetune] e241, train_loss:0.0118, r2:0.988, val_loss:0.2788, val_r2:0.725
[Finetune] e242, train_loss:0.0117, r2:0.988, val_loss:0.2789, val_r2:0.725
[Finetune] e243, train_loss:0.0117, r2:0.988, val_loss:0.2790, val_r2:0.725
[Finetune] e244, train_loss:0.0116, r2:0.989, val_loss:0.2791, val_r2:0.725
[Finetune] e245, train_loss:0.0116, r2:0.989, val_loss:0.2792, val_r2:0.725
[Finetune] e246, train_loss:0.0115, r2:0.989, val_loss:0.2793, val_r2:0.725
[Finetune] e247, train_loss:0.0115, r2:0.989, val_loss:0.2794, val_r2:0.725
[Finetune] e248, train_loss:0.0115, r2:0.989, val_loss:0.2795, val_r2:0.725
[Finetune] e249, train_loss:0.0114, r2:0.989, val_loss:0.2796, val_r2:0.725
[Finetune] e250, train_loss:0.0114, r2:0.989, val_loss:0.2797, val_r2:0.724
[Finetune] e251, train_loss:0.0113, r2:0.989, val_loss:0.2798, val_r2:0.724
[Finetune] e252, train_loss:0.0113, r2:0.989, val_loss:0.2799, val_r2:0.724
[Finetune] e253, train_loss:0.0112, r2:0.989, val_loss:0.2800, val_r2:0.724
[Finetune] e254, train_loss:0.0112, r2:0.989, val_loss:0.2801, val_r2:0.724
[Finetune] e255, train_loss:0.0111, r2:0.989, val_loss:0.2802, val_r2:0.724
[Finetune] e256, train_loss:0.0111, r2:0.989, val_loss:0.2803, val_r2:0.724
[Finetune] e257, train_loss:0.0110, r2:0.989, val_loss:0.2804, val_r2:0.724
[Finetune] e258, train_loss:0.0110, r2:0.989, val_loss:0.2805, val_r2:0.724
[Finetune] e259, train_loss:0.0110, r2:0.989, val_loss:0.2805, val_r2:0.724
[Finetune] e260, train_loss:0.0109, r2:0.989, val_loss:0.2806, val_r2:0.724

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```
[Finetune] e495, train_loss:0.0062, r2:0.994, val_loss:0.2819, val_r2:0.722
[Finetune] e496, train_loss:0.0062, r2:0.994, val_loss:0.2819, val_r2:0.722
[Finetune] e497, train_loss:0.0062, r2:0.994, val_loss:0.2819, val_r2:0.722
[Finetune] e498, train_loss:0.0062, r2:0.994, val_loss:0.2819, val_r2:0.722
[Finetune] e499, train_loss:0.0061, r2:0.994, val_loss:0.2819, val_r2:0.722
```

1.3 train_balanced_sampling.py

```
e0, train_loss:0.8055, r2:-0.042, val_loss:1.0358, val_r2:-0.020
Model successfully saved
e1, train_loss:0.7944, r2:-0.015, val_loss:1.0092, val_r2:0.006
Model successfully saved
e2, train_loss:0.7723, r2:0.011, val_loss:0.9835, val_r2:0.031
Model successfully saved
e3, train_loss:0.7422, r2:0.036, val_loss:0.9583, val_r2:0.056
Model successfully saved
e4, train_loss:0.7117, r2:0.061, val_loss:0.9332, val_r2:0.081
Model successfully saved
e5, train_loss:0.6932, r2:0.085, val_loss:0.9083, val_r2:0.105
Model successfully saved
e6, train_loss:0.6792, r2:0.109, val_loss:0.8832, val_r2:0.130
Model successfully saved
e7, train_loss:0.6606, r2:0.133, val_loss:0.8575, val_r2:0.155
Model successfully saved
e8, train_loss:0.6516, r2:0.157, val_loss:0.8313, val_r2:0.181
Model successfully saved
e9, train_loss:0.6123, r2:0.183, val_loss:0.8045, val_r2:0.207
Model successfully saved
e10, train_loss:0.5965, r2:0.208, val_loss:0.7775, val_r2:0.234
Model successfully saved
e11, train_loss:0.5905, r2:0.234, val_loss:0.7506, val_r2:0.261
Model successfully saved
e12, train_loss:0.5510, r2:0.261, val_loss:0.7238, val_r2:0.287
Model successfully saved
e13, train_loss:0.5356, r2:0.288, val_loss:0.6971, val_r2:0.313
Model successfully saved
e14, train_loss:0.5057, r2:0.316, val_loss:0.6699, val_r2:0.340
Model successfully saved
e15, train_loss:0.4887, r2:0.345, val_loss:0.6426, val_r2:0.367
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Model successfully saved
e16, train_loss:0.4641, r2:0.375, val_loss:0.6152, val_r2:0.394
Model successfully saved
e17, train_loss:0.4508, r2:0.405, val_loss:0.5880, val_r2:0.421
Model successfully saved
e18, train_loss:0.4255, r2:0.435, val_loss:0.5609, val_r2:0.447
Model successfully saved
e19, train_loss:0.4225, r2:0.465, val_loss:0.5333, val_r2:0.475
Model successfully saved
e20, train_loss:0.3886, r2:0.496, val_loss:0.5042, val_r2:0.503
Model successfully saved
e21, train_loss:0.3719, r2:0.526, val_loss:0.4739, val_r2:0.533
Model successfully saved
e22, train_loss:0.3692, r2:0.557, val_loss:0.4438, val_r2:0.563
Model successfully saved
e23, train_loss:0.3765, r2:0.586, val_loss:0.4167, val_r2:0.590
Model successfully saved
e24, train_loss:0.3476, r2:0.613, val_loss:0.3923, val_r2:0.614
Model successfully saved
e25, train_loss:0.3467, r2:0.638, val_loss:0.3714, val_r2:0.634
Model successfully saved
e26, train_loss:0.3334, r2:0.660, val_loss:0.3552, val_r2:0.650
Model successfully saved
e27, train_loss:0.3232, r2:0.680, val_loss:0.3421, val_r2:0.663
Model successfully saved
e28, train_loss:0.3174, r2:0.697, val_loss:0.3310, val_r2:0.674
Model successfully saved
e29, train_loss:0.2952, r2:0.712, val_loss:0.3201, val_r2:0.685
Model successfully saved
e30, train_loss:0.3189, r2:0.726, val_loss:0.3092, val_r2:0.695
Model successfully saved
e31, train_loss:0.2961, r2:0.739, val_loss:0.2972, val_r2:0.707
Model successfully saved
e32, train_loss:0.2832, r2:0.752, val_loss:0.2855, val_r2:0.719
Model successfully saved
e33, train_loss:0.2402, r2:0.764, val_loss:0.2735, val_r2:0.731
Model successfully saved
e34, train_loss:0.2677, r2:0.776, val_loss:0.2617, val_r2:0.742
Model successfully saved

e35, train_loss:0.2444, r2:0.788, val_loss:0.2506, val_r2:0.753
Model successfully saved

e36, train_loss:0.2434, r2:0.799, val_loss:0.2415, val_r2:0.762
Model successfully saved

e37, train_loss:0.2142, r2:0.810, val_loss:0.2333, val_r2:0.770
Model successfully saved

e38, train_loss:0.2091, r2:0.820, val_loss:0.2253, val_r2:0.778
Model successfully saved

e39, train_loss:0.2125, r2:0.830, val_loss:0.2182, val_r2:0.785
Model successfully saved

e40, train_loss:0.2003, r2:0.839, val_loss:0.2107, val_r2:0.792
Model successfully saved

e41, train_loss:0.2012, r2:0.848, val_loss:0.2042, val_r2:0.799
Model successfully saved

e42, train_loss:0.2032, r2:0.856, val_loss:0.1985, val_r2:0.804
Model successfully saved

e43, train_loss:0.2060, r2:0.864, val_loss:0.1933, val_r2:0.810
Model successfully saved

e44, train_loss:0.1908, r2:0.871, val_loss:0.1877, val_r2:0.815
Model successfully saved

e45, train_loss:0.1755, r2:0.878, val_loss:0.1821, val_r2:0.821
Model successfully saved

e46, train_loss:0.1780, r2:0.884, val_loss:0.1774, val_r2:0.825
Model successfully saved

e47, train_loss:0.1931, r2:0.890, val_loss:0.1745, val_r2:0.828
Model successfully saved

e48, train_loss:0.1560, r2:0.895, val_loss:0.1711, val_r2:0.831
Model successfully saved

e49, train_loss:0.1629, r2:0.900, val_loss:0.1684, val_r2:0.834
Model successfully saved

e50, train_loss:0.1675, r2:0.905, val_loss:0.1666, val_r2:0.836
Model successfully saved

e51, train_loss:0.1564, r2:0.909, val_loss:0.1652, val_r2:0.837
Model successfully saved

e52, train_loss:0.1627, r2:0.913, val_loss:0.1637, val_r2:0.839
Model successfully saved

e53, train_loss:0.1634, r2:0.917, val_loss:0.1631, val_r2:0.839
Model successfully saved

e54, train_loss:0.1631, r2:0.921, val_loss:0.1634, val_r2:0.839

e55, train_loss:0.1566, r2:0.924, val_loss:0.1641, val_r2:0.838
e56, train_loss:0.1529, r2:0.927, val_loss:0.1645, val_r2:0.838
e57, train_loss:0.1561, r2:0.930, val_loss:0.1637, val_r2:0.839
e58, train_loss:0.1610, r2:0.934, val_loss:0.1627, val_r2:0.840
Model successfully saved
e59, train_loss:0.1368, r2:0.936, val_loss:0.1611, val_r2:0.841
Model successfully saved
e60, train_loss:0.1570, r2:0.939, val_loss:0.1590, val_r2:0.843
Model successfully saved
e61, train_loss:0.1521, r2:0.942, val_loss:0.1577, val_r2:0.845
Model successfully saved
e62, train_loss:0.1529, r2:0.944, val_loss:0.1566, val_r2:0.846
Model successfully saved
e63, train_loss:0.1394, r2:0.946, val_loss:0.1562, val_r2:0.846
Model successfully saved
e64, train_loss:0.1389, r2:0.948, val_loss:0.1565, val_r2:0.846
e65, train_loss:0.1276, r2:0.950, val_loss:0.1555, val_r2:0.847
Model successfully saved
e66, train_loss:0.1331, r2:0.951, val_loss:0.1541, val_r2:0.848
Model successfully saved
e67, train_loss:0.1536, r2:0.953, val_loss:0.1542, val_r2:0.848
e68, train_loss:0.1430, r2:0.954, val_loss:0.1543, val_r2:0.848
e69, train_loss:0.1313, r2:0.955, val_loss:0.1537, val_r2:0.849
Model successfully saved
e70, train_loss:0.1462, r2:0.956, val_loss:0.1537, val_r2:0.849
e71, train_loss:0.1233, r2:0.957, val_loss:0.1523, val_r2:0.850
Model successfully saved
e72, train_loss:0.1205, r2:0.958, val_loss:0.1492, val_r2:0.853
Model successfully saved
e73, train_loss:0.1343, r2:0.960, val_loss:0.1466, val_r2:0.856
Model successfully saved
e74, train_loss:0.1309, r2:0.961, val_loss:0.1447, val_r2:0.857
Model successfully saved
e75, train_loss:0.1228, r2:0.963, val_loss:0.1438, val_r2:0.858
Model successfully saved
e76, train_loss:0.1192, r2:0.964, val_loss:0.1438, val_r2:0.858
e77, train_loss:0.1186, r2:0.965, val_loss:0.1444, val_r2:0.858
e78, train_loss:0.1325, r2:0.966, val_loss:0.1463, val_r2:0.856
e79, train_loss:0.1284, r2:0.966, val_loss:0.1489, val_r2:0.853

e80, train_loss:0.1256, r2:0.966, val_loss:0.1517, val_r2:0.851
e81, train_loss:0.1259, r2:0.967, val_loss:0.1524, val_r2:0.850
e82, train_loss:0.1240, r2:0.968, val_loss:0.1524, val_r2:0.850
e83, train_loss:0.1320, r2:0.969, val_loss:0.1519, val_r2:0.850
e84, train_loss:0.1195, r2:0.970, val_loss:0.1518, val_r2:0.850
e85, train_loss:0.1148, r2:0.971, val_loss:0.1521, val_r2:0.850
e86, train_loss:0.1221, r2:0.971, val_loss:0.1525, val_r2:0.850
e87, train_loss:0.1138, r2:0.972, val_loss:0.1528, val_r2:0.849
e88, train_loss:0.1373, r2:0.973, val_loss:0.1543, val_r2:0.848
e89, train_loss:0.1243, r2:0.973, val_loss:0.1556, val_r2:0.847
e90, train_loss:0.1151, r2:0.973, val_loss:0.1559, val_r2:0.846
e91, train_loss:0.1150, r2:0.974, val_loss:0.1563, val_r2:0.846
e92, train_loss:0.1200, r2:0.975, val_loss:0.1572, val_r2:0.845
e93, train_loss:0.1156, r2:0.976, val_loss:0.1582, val_r2:0.844
e94, train_loss:0.1196, r2:0.976, val_loss:0.1595, val_r2:0.843
e95, train_loss:0.1129, r2:0.977, val_loss:0.1604, val_r2:0.842
e96, train_loss:0.1122, r2:0.977, val_loss:0.1613, val_r2:0.841
e97, train_loss:0.1123, r2:0.977, val_loss:0.1616, val_r2:0.841
e98, train_loss:0.1139, r2:0.978, val_loss:0.1619, val_r2:0.840
e99, train_loss:0.1148, r2:0.978, val_loss:0.1619, val_r2:0.841
e100, train_loss:0.1051, r2:0.978, val_loss:0.1615, val_r2:0.841
e101, train_loss:0.1069, r2:0.979, val_loss:0.1610, val_r2:0.841
e102, train_loss:0.1088, r2:0.980, val_loss:0.1610, val_r2:0.841
e103, train_loss:0.1132, r2:0.980, val_loss:0.1617, val_r2:0.841
e104, train_loss:0.1121, r2:0.980, val_loss:0.1621, val_r2:0.840
e105, train_loss:0.1067, r2:0.981, val_loss:0.1622, val_r2:0.840
e106, train_loss:0.0985, r2:0.981, val_loss:0.1624, val_r2:0.840
e107, train_loss:0.0930, r2:0.982, val_loss:0.1627, val_r2:0.840
e108, train_loss:0.1155, r2:0.982, val_loss:0.1641, val_r2:0.838
e109, train_loss:0.1027, r2:0.982, val_loss:0.1661, val_r2:0.836
e110, train_loss:0.1006, r2:0.982, val_loss:0.1690, val_r2:0.834
e111, train_loss:0.1074, r2:0.982, val_loss:0.1718, val_r2:0.831
e112, train_loss:0.1118, r2:0.983, val_loss:0.1735, val_r2:0.829
e113, train_loss:0.1135, r2:0.983, val_loss:0.1730, val_r2:0.830
e114, train_loss:0.1012, r2:0.984, val_loss:0.1722, val_r2:0.830
e115, train_loss:0.1011, r2:0.984, val_loss:0.1720, val_r2:0.831
e116, train_loss:0.1079, r2:0.984, val_loss:0.1708, val_r2:0.832
e117, train_loss:0.1038, r2:0.984, val_loss:0.1701, val_r2:0.832
e118, train_loss:0.0871, r2:0.985, val_loss:0.1712, val_r2:0.831

e119, train_loss:0.0989, r2:0.985, val_loss:0.1727, val_r2:0.830
e120, train_loss:0.1042, r2:0.985, val_loss:0.1732, val_r2:0.829
e121, train_loss:0.1163, r2:0.985, val_loss:0.1732, val_r2:0.829
e122, train_loss:0.1001, r2:0.986, val_loss:0.1728, val_r2:0.830
e123, train_loss:0.0967, r2:0.986, val_loss:0.1732, val_r2:0.829
e124, train_loss:0.0990, r2:0.986, val_loss:0.1732, val_r2:0.829
e125, train_loss:0.0959, r2:0.987, val_loss:0.1731, val_r2:0.829
e126, train_loss:0.0939, r2:0.987, val_loss:0.1732, val_r2:0.829
e127, train_loss:0.1140, r2:0.987, val_loss:0.1739, val_r2:0.829
e128, train_loss:0.0982, r2:0.987, val_loss:0.1749, val_r2:0.828
e129, train_loss:0.1145, r2:0.987, val_loss:0.1762, val_r2:0.826
e130, train_loss:0.0998, r2:0.988, val_loss:0.1771, val_r2:0.826
e131, train_loss:0.0954, r2:0.988, val_loss:0.1783, val_r2:0.824
e132, train_loss:0.1072, r2:0.989, val_loss:0.1793, val_r2:0.823
e133, train_loss:0.0982, r2:0.989, val_loss:0.1795, val_r2:0.823
e134, train_loss:0.1014, r2:0.989, val_loss:0.1798, val_r2:0.823
e135, train_loss:0.0952, r2:0.989, val_loss:0.1806, val_r2:0.822
e136, train_loss:0.0918, r2:0.989, val_loss:0.1811, val_r2:0.822
e137, train_loss:0.0981, r2:0.989, val_loss:0.1820, val_r2:0.821
e138, train_loss:0.0996, r2:0.989, val_loss:0.1833, val_r2:0.819
e139, train_loss:0.0947, r2:0.990, val_loss:0.1845, val_r2:0.818
e140, train_loss:0.1017, r2:0.990, val_loss:0.1855, val_r2:0.817
e141, train_loss:0.0966, r2:0.990, val_loss:0.1861, val_r2:0.817
e142, train_loss:0.0963, r2:0.990, val_loss:0.1872, val_r2:0.816
e143, train_loss:0.0967, r2:0.990, val_loss:0.1882, val_r2:0.815
e144, train_loss:0.0912, r2:0.991, val_loss:0.1875, val_r2:0.815
e145, train_loss:0.0960, r2:0.991, val_loss:0.1867, val_r2:0.816
e146, train_loss:0.0964, r2:0.991, val_loss:0.1861, val_r2:0.817
e147, train_loss:0.1061, r2:0.991, val_loss:0.1853, val_r2:0.818
e148, train_loss:0.0932, r2:0.991, val_loss:0.1841, val_r2:0.819
e149, train_loss:0.1037, r2:0.992, val_loss:0.1834, val_r2:0.819
e150, train_loss:0.0999, r2:0.992, val_loss:0.1837, val_r2:0.819
e151, train_loss:0.0966, r2:0.992, val_loss:0.1844, val_r2:0.818
e152, train_loss:0.0976, r2:0.992, val_loss:0.1855, val_r2:0.817
e153, train_loss:0.0931, r2:0.992, val_loss:0.1863, val_r2:0.816
e154, train_loss:0.0951, r2:0.992, val_loss:0.1864, val_r2:0.816
e155, train_loss:0.1014, r2:0.992, val_loss:0.1863, val_r2:0.816
e156, train_loss:0.1075, r2:0.992, val_loss:0.1865, val_r2:0.816
e157, train_loss:0.0956, r2:0.992, val_loss:0.1867, val_r2:0.816

e158, train_loss:0.0976, r2:0.992, val_loss:0.1870, val_r2:0.816
e159, train_loss:0.1004, r2:0.992, val_loss:0.1881, val_r2:0.815
e160, train_loss:0.0901, r2:0.992, val_loss:0.1882, val_r2:0.815
e161, train_loss:0.0952, r2:0.993, val_loss:0.1887, val_r2:0.814
e162, train_loss:0.0952, r2:0.993, val_loss:0.1893, val_r2:0.813
e163, train_loss:0.1012, r2:0.993, val_loss:0.1901, val_r2:0.813
e164, train_loss:0.1019, r2:0.993, val_loss:0.1914, val_r2:0.811
e165, train_loss:0.0999, r2:0.993, val_loss:0.1914, val_r2:0.811
e166, train_loss:0.0967, r2:0.993, val_loss:0.1903, val_r2:0.813
e167, train_loss:0.0955, r2:0.993, val_loss:0.1888, val_r2:0.814
e168, train_loss:0.0904, r2:0.993, val_loss:0.1874, val_r2:0.815
e169, train_loss:0.0925, r2:0.993, val_loss:0.1859, val_r2:0.817
e170, train_loss:0.0965, r2:0.993, val_loss:0.1849, val_r2:0.818
e171, train_loss:0.1000, r2:0.993, val_loss:0.1859, val_r2:0.817
e172, train_loss:0.0945, r2:0.993, val_loss:0.1874, val_r2:0.815
e173, train_loss:0.0922, r2:0.993, val_loss:0.1887, val_r2:0.814
e174, train_loss:0.0906, r2:0.993, val_loss:0.1894, val_r2:0.813
e175, train_loss:0.0882, r2:0.993, val_loss:0.1911, val_r2:0.812
e176, train_loss:0.0989, r2:0.993, val_loss:0.1923, val_r2:0.811
e177, train_loss:0.1008, r2:0.993, val_loss:0.1931, val_r2:0.810
e178, train_loss:0.0916, r2:0.994, val_loss:0.1948, val_r2:0.808
e179, train_loss:0.0792, r2:0.993, val_loss:0.1957, val_r2:0.807
e180, train_loss:0.0919, r2:0.993, val_loss:0.1955, val_r2:0.807
e181, train_loss:0.0926, r2:0.993, val_loss:0.1939, val_r2:0.809
e182, train_loss:0.0928, r2:0.993, val_loss:0.1921, val_r2:0.811
e183, train_loss:0.0913, r2:0.994, val_loss:0.1891, val_r2:0.814
e184, train_loss:0.1008, r2:0.994, val_loss:0.1860, val_r2:0.817
e185, train_loss:0.1014, r2:0.994, val_loss:0.1842, val_r2:0.819
e186, train_loss:0.1029, r2:0.994, val_loss:0.1831, val_r2:0.820
e187, train_loss:0.0890, r2:0.994, val_loss:0.1814, val_r2:0.821
e188, train_loss:0.0929, r2:0.994, val_loss:0.1808, val_r2:0.822
e189, train_loss:0.1019, r2:0.994, val_loss:0.1819, val_r2:0.821
e190, train_loss:0.0890, r2:0.994, val_loss:0.1837, val_r2:0.819
e191, train_loss:0.0925, r2:0.994, val_loss:0.1860, val_r2:0.817
e192, train_loss:0.0952, r2:0.994, val_loss:0.1876, val_r2:0.815
e193, train_loss:0.0925, r2:0.994, val_loss:0.1891, val_r2:0.814
e194, train_loss:0.0929, r2:0.994, val_loss:0.1911, val_r2:0.812
e195, train_loss:0.0842, r2:0.994, val_loss:0.1924, val_r2:0.810
e196, train_loss:0.0986, r2:0.994, val_loss:0.1915, val_r2:0.811

e197, train_loss:0.0969, r2:0.994, val_loss:0.1909, val_r2:0.812
e198, train_loss:0.0972, r2:0.994, val_loss:0.1895, val_r2:0.813
e199, train_loss:0.0952, r2:0.994, val_loss:0.1867, val_r2:0.816
e200, train_loss:0.0896, r2:0.994, val_loss:0.1842, val_r2:0.819
e201, train_loss:0.0998, r2:0.994, val_loss:0.1817, val_r2:0.821
e202, train_loss:0.1016, r2:0.995, val_loss:0.1821, val_r2:0.821
e203, train_loss:0.0898, r2:0.994, val_loss:0.1834, val_r2:0.819
e204, train_loss:0.0960, r2:0.994, val_loss:0.1844, val_r2:0.818
e205, train_loss:0.0954, r2:0.995, val_loss:0.1866, val_r2:0.816
e206, train_loss:0.0993, r2:0.995, val_loss:0.1886, val_r2:0.814
e207, train_loss:0.0985, r2:0.995, val_loss:0.1899, val_r2:0.813
e208, train_loss:0.0897, r2:0.995, val_loss:0.1895, val_r2:0.813
e209, train_loss:0.0950, r2:0.995, val_loss:0.1887, val_r2:0.814
e210, train_loss:0.0942, r2:0.995, val_loss:0.1890, val_r2:0.814
e211, train_loss:0.0909, r2:0.995, val_loss:0.1904, val_r2:0.812
e212, train_loss:0.0958, r2:0.995, val_loss:0.1927, val_r2:0.810
e213, train_loss:0.0974, r2:0.995, val_loss:0.1945, val_r2:0.808
e214, train_loss:0.0892, r2:0.995, val_loss:0.1955, val_r2:0.807
e215, train_loss:0.0904, r2:0.995, val_loss:0.1953, val_r2:0.808
e216, train_loss:0.0948, r2:0.995, val_loss:0.1937, val_r2:0.809
e217, train_loss:0.0946, r2:0.995, val_loss:0.1925, val_r2:0.810
e218, train_loss:0.0880, r2:0.995, val_loss:0.1910, val_r2:0.812
e219, train_loss:0.1010, r2:0.995, val_loss:0.1887, val_r2:0.814
e220, train_loss:0.0834, r2:0.995, val_loss:0.1860, val_r2:0.817
e221, train_loss:0.0923, r2:0.995, val_loss:0.1851, val_r2:0.818
e222, train_loss:0.0833, r2:0.994, val_loss:0.1855, val_r2:0.817
e223, train_loss:0.0909, r2:0.995, val_loss:0.1871, val_r2:0.816
e224, train_loss:0.0907, r2:0.995, val_loss:0.1899, val_r2:0.813
e225, train_loss:0.0918, r2:0.995, val_loss:0.1939, val_r2:0.809
e226, train_loss:0.0973, r2:0.995, val_loss:0.1950, val_r2:0.808
e227, train_loss:0.0979, r2:0.995, val_loss:0.1958, val_r2:0.807
e228, train_loss:0.0942, r2:0.995, val_loss:0.1961, val_r2:0.807
e229, train_loss:0.0939, r2:0.995, val_loss:0.1949, val_r2:0.808
e230, train_loss:0.0955, r2:0.995, val_loss:0.1926, val_r2:0.810
e231, train_loss:0.0969, r2:0.995, val_loss:0.1901, val_r2:0.813
e232, train_loss:0.0891, r2:0.995, val_loss:0.1883, val_r2:0.815
e233, train_loss:0.1116, r2:0.995, val_loss:0.1876, val_r2:0.815
e234, train_loss:0.1000, r2:0.995, val_loss:0.1879, val_r2:0.815
e235, train_loss:0.0961, r2:0.995, val_loss:0.1874, val_r2:0.815

e236, train_loss:0.0971, r2:0.995, val_loss:0.1877, val_r2:0.815
e237, train_loss:0.0928, r2:0.995, val_loss:0.1883, val_r2:0.815
e238, train_loss:0.0802, r2:0.995, val_loss:0.1885, val_r2:0.814
e239, train_loss:0.0940, r2:0.995, val_loss:0.1894, val_r2:0.813
e240, train_loss:0.0798, r2:0.995, val_loss:0.1901, val_r2:0.813
e241, train_loss:0.0852, r2:0.995, val_loss:0.1913, val_r2:0.812
e242, train_loss:0.0873, r2:0.995, val_loss:0.1929, val_r2:0.810
e243, train_loss:0.0926, r2:0.995, val_loss:0.1945, val_r2:0.808
e244, train_loss:0.0864, r2:0.995, val_loss:0.1942, val_r2:0.809
e245, train_loss:0.0976, r2:0.995, val_loss:0.1921, val_r2:0.811
e246, train_loss:0.1005, r2:0.995, val_loss:0.1880, val_r2:0.815
e247, train_loss:0.0974, r2:0.995, val_loss:0.1837, val_r2:0.819
e248, train_loss:0.0926, r2:0.996, val_loss:0.1802, val_r2:0.822
e249, train_loss:0.0906, r2:0.996, val_loss:0.1790, val_r2:0.824
e250, train_loss:0.0968, r2:0.995, val_loss:0.1798, val_r2:0.823
e251, train_loss:0.0893, r2:0.995, val_loss:0.1815, val_r2:0.821
e252, train_loss:0.0839, r2:0.995, val_loss:0.1840, val_r2:0.819
e253, train_loss:0.0876, r2:0.996, val_loss:0.1863, val_r2:0.816
e254, train_loss:0.0960, r2:0.995, val_loss:0.1882, val_r2:0.815
e255, train_loss:0.0934, r2:0.996, val_loss:0.1901, val_r2:0.813
e256, train_loss:0.0882, r2:0.996, val_loss:0.1928, val_r2:0.810
e257, train_loss:0.0889, r2:0.995, val_loss:0.1933, val_r2:0.810
e258, train_loss:0.0904, r2:0.996, val_loss:0.1931, val_r2:0.810
e259, train_loss:0.0957, r2:0.996, val_loss:0.1920, val_r2:0.811
e260, train_loss:0.1039, r2:0.996, val_loss:0.1906, val_r2:0.812
e261, train_loss:0.0891, r2:0.996, val_loss:0.1891, val_r2:0.814
e262, train_loss:0.0930, r2:0.995, val_loss:0.1864, val_r2:0.816
e263, train_loss:0.0922, r2:0.995, val_loss:0.1816, val_r2:0.821
e264, train_loss:0.0977, r2:0.996, val_loss:0.1790, val_r2:0.824
e265, train_loss:0.0922, r2:0.995, val_loss:0.1780, val_r2:0.825
e266, train_loss:0.0901, r2:0.995, val_loss:0.1784, val_r2:0.824
e267, train_loss:0.0953, r2:0.995, val_loss:0.1814, val_r2:0.821
e268, train_loss:0.0928, r2:0.995, val_loss:0.1869, val_r2:0.816
e269, train_loss:0.0916, r2:0.995, val_loss:0.1902, val_r2:0.813
e270, train_loss:0.0893, r2:0.995, val_loss:0.1920, val_r2:0.811
e271, train_loss:0.1028, r2:0.995, val_loss:0.1929, val_r2:0.810
e272, train_loss:0.1004, r2:0.995, val_loss:0.1904, val_r2:0.812
e273, train_loss:0.0965, r2:0.995, val_loss:0.1869, val_r2:0.816
e274, train_loss:0.0900, r2:0.996, val_loss:0.1863, val_r2:0.816

e275, train_loss:0.0972, r2:0.994, val_loss:0.1861, val_r2:0.817
e276, train_loss:0.0972, r2:0.994, val_loss:0.1830, val_r2:0.820
e277, train_loss:0.0953, r2:0.996, val_loss:0.1826, val_r2:0.820
e278, train_loss:0.0932, r2:0.994, val_loss:0.1840, val_r2:0.819
e279, train_loss:0.0914, r2:0.993, val_loss:0.1837, val_r2:0.819
e280, train_loss:0.1041, r2:0.995, val_loss:0.1876, val_r2:0.815
e281, train_loss:0.0953, r2:0.996, val_loss:0.1914, val_r2:0.811
e282, train_loss:0.0902, r2:0.995, val_loss:0.1915, val_r2:0.811
e283, train_loss:0.0975, r2:0.996, val_loss:0.1923, val_r2:0.811
e284, train_loss:0.1002, r2:0.996, val_loss:0.1942, val_r2:0.809
e285, train_loss:0.0898, r2:0.995, val_loss:0.1928, val_r2:0.810
e286, train_loss:0.0944, r2:0.996, val_loss:0.1915, val_r2:0.811
e287, train_loss:0.0784, r2:0.996, val_loss:0.1912, val_r2:0.812
e288, train_loss:0.0907, r2:0.995, val_loss:0.1884, val_r2:0.814
e289, train_loss:0.0875, r2:0.995, val_loss:0.1846, val_r2:0.818
e290, train_loss:0.0914, r2:0.996, val_loss:0.1834, val_r2:0.819
e291, train_loss:0.0842, r2:0.995, val_loss:0.1832, val_r2:0.819
e292, train_loss:0.0959, r2:0.995, val_loss:0.1825, val_r2:0.820
e293, train_loss:0.0879, r2:0.996, val_loss:0.1841, val_r2:0.819
e294, train_loss:0.1005, r2:0.996, val_loss:0.1883, val_r2:0.815
e295, train_loss:0.1031, r2:0.995, val_loss:0.1899, val_r2:0.813
e296, train_loss:0.0888, r2:0.996, val_loss:0.1910, val_r2:0.812
e297, train_loss:0.0897, r2:0.996, val_loss:0.1913, val_r2:0.812
e298, train_loss:0.0763, r2:0.995, val_loss:0.1903, val_r2:0.812
e299, train_loss:0.1023, r2:0.995, val_loss:0.1875, val_r2:0.815
e300, train_loss:0.0818, r2:0.996, val_loss:0.1850, val_r2:0.818
e301, train_loss:0.0951, r2:0.996, val_loss:0.1845, val_r2:0.818
e302, train_loss:0.0954, r2:0.995, val_loss:0.1868, val_r2:0.816
e303, train_loss:0.0949, r2:0.995, val_loss:0.1885, val_r2:0.814
e304, train_loss:0.0974, r2:0.995, val_loss:0.1882, val_r2:0.815
e305, train_loss:0.0943, r2:0.996, val_loss:0.1879, val_r2:0.815
e306, train_loss:0.0952, r2:0.995, val_loss:0.1872, val_r2:0.816
e307, train_loss:0.0865, r2:0.994, val_loss:0.1848, val_r2:0.818
e308, train_loss:0.0920, r2:0.996, val_loss:0.1839, val_r2:0.819
e309, train_loss:0.1089, r2:0.996, val_loss:0.1869, val_r2:0.816
e310, train_loss:0.0959, r2:0.995, val_loss:0.1903, val_r2:0.813
e311, train_loss:0.0925, r2:0.995, val_loss:0.1911, val_r2:0.812
e312, train_loss:0.0863, r2:0.996, val_loss:0.1916, val_r2:0.811
e313, train_loss:0.0914, r2:0.996, val_loss:0.1916, val_r2:0.811

e314, train_loss:0.0874, r2:0.995, val_loss:0.1891, val_r2:0.814
e315, train_loss:0.0859, r2:0.996, val_loss:0.1871, val_r2:0.816
e316, train_loss:0.0882, r2:0.996, val_loss:0.1858, val_r2:0.817
e317, train_loss:0.0821, r2:0.996, val_loss:0.1850, val_r2:0.818
e318, train_loss:0.0859, r2:0.996, val_loss:0.1844, val_r2:0.818
e319, train_loss:0.0858, r2:0.996, val_loss:0.1856, val_r2:0.817
e320, train_loss:0.0938, r2:0.996, val_loss:0.1882, val_r2:0.815
e321, train_loss:0.0844, r2:0.995, val_loss:0.1892, val_r2:0.814
e322, train_loss:0.0850, r2:0.996, val_loss:0.1908, val_r2:0.812
e323, train_loss:0.0848, r2:0.996, val_loss:0.1927, val_r2:0.810
e324, train_loss:0.0836, r2:0.996, val_loss:0.1921, val_r2:0.811
e325, train_loss:0.0923, r2:0.996, val_loss:0.1905, val_r2:0.812
e326, train_loss:0.1023, r2:0.996, val_loss:0.1896, val_r2:0.813
e327, train_loss:0.0818, r2:0.996, val_loss:0.1890, val_r2:0.814
e328, train_loss:0.0928, r2:0.996, val_loss:0.1888, val_r2:0.814
e329, train_loss:0.0923, r2:0.996, val_loss:0.1882, val_r2:0.815
e330, train_loss:0.0942, r2:0.996, val_loss:0.1880, val_r2:0.815
e331, train_loss:0.0922, r2:0.996, val_loss:0.1881, val_r2:0.815
e332, train_loss:0.0979, r2:0.997, val_loss:0.1888, val_r2:0.814
e333, train_loss:0.0898, r2:0.996, val_loss:0.1894, val_r2:0.813
e334, train_loss:0.0864, r2:0.996, val_loss:0.1894, val_r2:0.813
e335, train_loss:0.0926, r2:0.996, val_loss:0.1891, val_r2:0.814
e336, train_loss:0.0970, r2:0.996, val_loss:0.1892, val_r2:0.814
e337, train_loss:0.0857, r2:0.996, val_loss:0.1894, val_r2:0.813
e338, train_loss:0.0914, r2:0.996, val_loss:0.1900, val_r2:0.813
e339, train_loss:0.0847, r2:0.996, val_loss:0.1904, val_r2:0.812
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e341, train_loss:0.0812, r2:0.996, val_loss:0.1901, val_r2:0.813
e342, train_loss:0.0851, r2:0.996, val_loss:0.1892, val_r2:0.814
e343, train_loss:0.0918, r2:0.996, val_loss:0.1890, val_r2:0.814
e344, train_loss:0.0920, r2:0.997, val_loss:0.1899, val_r2:0.813
e345, train_loss:0.0950, r2:0.996, val_loss:0.1924, val_r2:0.810
e346, train_loss:0.0860, r2:0.996, val_loss:0.1952, val_r2:0.808
e347, train_loss:0.0914, r2:0.996, val_loss:0.1971, val_r2:0.806
e348, train_loss:0.0806, r2:0.996, val_loss:0.1988, val_r2:0.804
e349, train_loss:0.0902, r2:0.996, val_loss:0.1997, val_r2:0.803
e350, train_loss:0.0860, r2:0.996, val_loss:0.2010, val_r2:0.802
e351, train_loss:0.1001, r2:0.997, val_loss:0.2021, val_r2:0.801
e352, train_loss:0.0876, r2:0.996, val_loss:0.2009, val_r2:0.802

e353, train_loss:0.0876, r2:0.996, val_loss:0.1981, val_r2:0.805
e354, train_loss:0.1015, r2:0.997, val_loss:0.1954, val_r2:0.807
e355, train_loss:0.0950, r2:0.996, val_loss:0.1932, val_r2:0.810
e356, train_loss:0.0797, r2:0.997, val_loss:0.1913, val_r2:0.812
e357, train_loss:0.0930, r2:0.997, val_loss:0.1903, val_r2:0.813
e358, train_loss:0.0961, r2:0.997, val_loss:0.1897, val_r2:0.813
e359, train_loss:0.0859, r2:0.997, val_loss:0.1904, val_r2:0.812
e360, train_loss:0.0963, r2:0.997, val_loss:0.1918, val_r2:0.811
e361, train_loss:0.0823, r2:0.997, val_loss:0.1917, val_r2:0.811
e362, train_loss:0.0889, r2:0.997, val_loss:0.1917, val_r2:0.811
e363, train_loss:0.0956, r2:0.996, val_loss:0.1916, val_r2:0.811
e364, train_loss:0.0967, r2:0.997, val_loss:0.1928, val_r2:0.810
e365, train_loss:0.0930, r2:0.996, val_loss:0.1969, val_r2:0.806
e366, train_loss:0.0893, r2:0.995, val_loss:0.1968, val_r2:0.806
e367, train_loss:0.0977, r2:0.996, val_loss:0.1937, val_r2:0.809
e368, train_loss:0.0998, r2:0.997, val_loss:0.1911, val_r2:0.812
e369, train_loss:0.0893, r2:0.996, val_loss:0.1883, val_r2:0.815
e370, train_loss:0.0966, r2:0.996, val_loss:0.1858, val_r2:0.817
e371, train_loss:0.0898, r2:0.996, val_loss:0.1853, val_r2:0.817
e372, train_loss:0.0905, r2:0.996, val_loss:0.1867, val_r2:0.816
e373, train_loss:0.0945, r2:0.996, val_loss:0.1889, val_r2:0.814
e374, train_loss:0.0907, r2:0.997, val_loss:0.1919, val_r2:0.811
e375, train_loss:0.1025, r2:0.997, val_loss:0.1954, val_r2:0.807
e376, train_loss:0.0777, r2:0.997, val_loss:0.1971, val_r2:0.806
e377, train_loss:0.0911, r2:0.996, val_loss:0.1961, val_r2:0.807
e378, train_loss:0.0854, r2:0.996, val_loss:0.1937, val_r2:0.809
e379, train_loss:0.0900, r2:0.997, val_loss:0.1902, val_r2:0.813
e380, train_loss:0.0891, r2:0.997, val_loss:0.1876, val_r2:0.815
e381, train_loss:0.0821, r2:0.997, val_loss:0.1866, val_r2:0.816
e382, train_loss:0.0866, r2:0.996, val_loss:0.1879, val_r2:0.815
e383, train_loss:0.0840, r2:0.997, val_loss:0.1905, val_r2:0.812
e384, train_loss:0.1006, r2:0.997, val_loss:0.1928, val_r2:0.810
e385, train_loss:0.0821, r2:0.997, val_loss:0.1949, val_r2:0.808
e386, train_loss:0.0920, r2:0.996, val_loss:0.1949, val_r2:0.808
e387, train_loss:0.0821, r2:0.997, val_loss:0.1939, val_r2:0.809
e388, train_loss:0.0925, r2:0.997, val_loss:0.1945, val_r2:0.808
e389, train_loss:0.0845, r2:0.997, val_loss:0.1960, val_r2:0.807
e390, train_loss:0.0812, r2:0.996, val_loss:0.1959, val_r2:0.807
e391, train_loss:0.0948, r2:0.997, val_loss:0.1953, val_r2:0.808

e392, train_loss:0.0822, r2:0.997, val_loss:0.1942, val_r2:0.809
e393, train_loss:0.0940, r2:0.996, val_loss:0.1924, val_r2:0.810
e394, train_loss:0.0940, r2:0.997, val_loss:0.1905, val_r2:0.812
e395, train_loss:0.0891, r2:0.997, val_loss:0.1905, val_r2:0.812
e396, train_loss:0.0890, r2:0.997, val_loss:0.1926, val_r2:0.810
e397, train_loss:0.0976, r2:0.996, val_loss:0.1912, val_r2:0.812
e398, train_loss:0.0879, r2:0.997, val_loss:0.1918, val_r2:0.811
e399, train_loss:0.0827, r2:0.996, val_loss:0.1942, val_r2:0.809
e400, train_loss:0.0917, r2:0.995, val_loss:0.1948, val_r2:0.808
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e402, train_loss:0.0875, r2:0.996, val_loss:0.2037, val_r2:0.799
e403, train_loss:0.0890, r2:0.995, val_loss:0.2033, val_r2:0.800
e404, train_loss:0.0885, r2:0.996, val_loss:0.2009, val_r2:0.802
e405, train_loss:0.0879, r2:0.997, val_loss:0.1988, val_r2:0.804
e406, train_loss:0.0920, r2:0.995, val_loss:0.1942, val_r2:0.809
e407, train_loss:0.0859, r2:0.996, val_loss:0.1892, val_r2:0.814
e408, train_loss:0.0868, r2:0.997, val_loss:0.1865, val_r2:0.816
e409, train_loss:0.0885, r2:0.996, val_loss:0.1840, val_r2:0.819
e410, train_loss:0.0923, r2:0.996, val_loss:0.1827, val_r2:0.820
e411, train_loss:0.0922, r2:0.996, val_loss:0.1848, val_r2:0.818
e412, train_loss:0.0900, r2:0.997, val_loss:0.1890, val_r2:0.814
e413, train_loss:0.0933, r2:0.996, val_loss:0.1922, val_r2:0.811
e414, train_loss:0.0914, r2:0.996, val_loss:0.1951, val_r2:0.808
e415, train_loss:0.0875, r2:0.997, val_loss:0.1963, val_r2:0.807
e416, train_loss:0.0887, r2:0.997, val_loss:0.1960, val_r2:0.807
e417, train_loss:0.0933, r2:0.997, val_loss:0.1965, val_r2:0.806
e418, train_loss:0.0912, r2:0.997, val_loss:0.1972, val_r2:0.806
e419, train_loss:0.0929, r2:0.997, val_loss:0.1974, val_r2:0.806
e420, train_loss:0.0922, r2:0.997, val_loss:0.1976, val_r2:0.805
e421, train_loss:0.0918, r2:0.997, val_loss:0.1960, val_r2:0.807
e422, train_loss:0.0866, r2:0.997, val_loss:0.1929, val_r2:0.810
e423, train_loss:0.0791, r2:0.997, val_loss:0.1898, val_r2:0.813
e424, train_loss:0.0909, r2:0.997, val_loss:0.1872, val_r2:0.816
e425, train_loss:0.0790, r2:0.997, val_loss:0.1858, val_r2:0.817
e426, train_loss:0.0808, r2:0.997, val_loss:0.1867, val_r2:0.816
e427, train_loss:0.0865, r2:0.997, val_loss:0.1897, val_r2:0.813
e428, train_loss:0.0722, r2:0.997, val_loss:0.1929, val_r2:0.810
e429, train_loss:0.0901, r2:0.997, val_loss:0.1955, val_r2:0.807
e430, train_loss:0.0970, r2:0.997, val_loss:0.1976, val_r2:0.805

e431, train_loss:0.0938, r2:0.997, val_loss:0.1987, val_r2:0.804
e432, train_loss:0.0875, r2:0.997, val_loss:0.1979, val_r2:0.805
e433, train_loss:0.0926, r2:0.997, val_loss:0.1962, val_r2:0.807
e434, train_loss:0.0823, r2:0.997, val_loss:0.1948, val_r2:0.808
e435, train_loss:0.0902, r2:0.997, val_loss:0.1946, val_r2:0.808
e436, train_loss:0.0867, r2:0.997, val_loss:0.1946, val_r2:0.808
e437, train_loss:0.0825, r2:0.997, val_loss:0.1937, val_r2:0.809
e438, train_loss:0.0918, r2:0.997, val_loss:0.1927, val_r2:0.810
e439, train_loss:0.0944, r2:0.997, val_loss:0.1926, val_r2:0.810
e440, train_loss:0.0952, r2:0.997, val_loss:0.1933, val_r2:0.810
e441, train_loss:0.0876, r2:0.997, val_loss:0.1926, val_r2:0.810
e442, train_loss:0.0851, r2:0.997, val_loss:0.1925, val_r2:0.810
e443, train_loss:0.0894, r2:0.997, val_loss:0.1941, val_r2:0.809
e444, train_loss:0.0866, r2:0.997, val_loss:0.1961, val_r2:0.807
e445, train_loss:0.0852, r2:0.997, val_loss:0.1979, val_r2:0.805
e446, train_loss:0.0778, r2:0.997, val_loss:0.1988, val_r2:0.804
e447, train_loss:0.0879, r2:0.997, val_loss:0.1988, val_r2:0.804
e448, train_loss:0.0896, r2:0.997, val_loss:0.1988, val_r2:0.804
e449, train_loss:0.0841, r2:0.997, val_loss:0.1978, val_r2:0.805
e450, train_loss:0.0756, r2:0.996, val_loss:0.1945, val_r2:0.808
e451, train_loss:0.0930, r2:0.997, val_loss:0.1909, val_r2:0.812
e452, train_loss:0.0871, r2:0.997, val_loss:0.1874, val_r2:0.815
e453, train_loss:0.0890, r2:0.997, val_loss:0.1853, val_r2:0.817
e454, train_loss:0.0921, r2:0.997, val_loss:0.1854, val_r2:0.817
e455, train_loss:0.0801, r2:0.997, val_loss:0.1872, val_r2:0.816
e456, train_loss:0.0888, r2:0.997, val_loss:0.1906, val_r2:0.812
e457, train_loss:0.0845, r2:0.997, val_loss:0.1952, val_r2:0.808
e458, train_loss:0.0899, r2:0.997, val_loss:0.1992, val_r2:0.804
e459, train_loss:0.0894, r2:0.997, val_loss:0.2019, val_r2:0.801
e460, train_loss:0.0865, r2:0.997, val_loss:0.2026, val_r2:0.800
e461, train_loss:0.0895, r2:0.997, val_loss:0.2002, val_r2:0.803
e462, train_loss:0.0897, r2:0.997, val_loss:0.1963, val_r2:0.807
e463, train_loss:0.0819, r2:0.997, val_loss:0.1909, val_r2:0.812
e464, train_loss:0.0937, r2:0.997, val_loss:0.1869, val_r2:0.816
e465, train_loss:0.0906, r2:0.997, val_loss:0.1844, val_r2:0.818
e466, train_loss:0.0889, r2:0.997, val_loss:0.1844, val_r2:0.818
e467, train_loss:0.0931, r2:0.997, val_loss:0.1863, val_r2:0.816
e468, train_loss:0.0824, r2:0.997, val_loss:0.1897, val_r2:0.813
e469, train_loss:0.1003, r2:0.997, val_loss:0.1940, val_r2:0.809

e470, train_loss:0.1060, r2:0.997, val_loss:0.1979, val_r2:0.805
e471, train_loss:0.0855, r2:0.997, val_loss:0.2000, val_r2:0.803
e472, train_loss:0.0829, r2:0.997, val_loss:0.2006, val_r2:0.802
e473, train_loss:0.0922, r2:0.997, val_loss:0.2009, val_r2:0.802
e474, train_loss:0.0985, r2:0.997, val_loss:0.2021, val_r2:0.801
e475, train_loss:0.0887, r2:0.997, val_loss:0.2004, val_r2:0.803
e476, train_loss:0.0869, r2:0.997, val_loss:0.1978, val_r2:0.805
e477, train_loss:0.0938, r2:0.997, val_loss:0.1939, val_r2:0.809
e478, train_loss:0.0801, r2:0.997, val_loss:0.1897, val_r2:0.813
e479, train_loss:0.0877, r2:0.997, val_loss:0.1859, val_r2:0.817
e480, train_loss:0.0920, r2:0.997, val_loss:0.1835, val_r2:0.819
e481, train_loss:0.0786, r2:0.997, val_loss:0.1836, val_r2:0.819
e482, train_loss:0.0888, r2:0.997, val_loss:0.1851, val_r2:0.818
e483, train_loss:0.0868, r2:0.997, val_loss:0.1885, val_r2:0.814
e484, train_loss:0.0871, r2:0.997, val_loss:0.1925, val_r2:0.810
e485, train_loss:0.0975, r2:0.997, val_loss:0.1958, val_r2:0.807
e486, train_loss:0.0880, r2:0.997, val_loss:0.1972, val_r2:0.806
e487, train_loss:0.0833, r2:0.997, val_loss:0.1962, val_r2:0.807
e488, train_loss:0.0905, r2:0.997, val_loss:0.1936, val_r2:0.809
e489, train_loss:0.0926, r2:0.997, val_loss:0.1907, val_r2:0.812
e490, train_loss:0.0852, r2:0.997, val_loss:0.1886, val_r2:0.814
e491, train_loss:0.0879, r2:0.997, val_loss:0.1879, val_r2:0.815
e492, train_loss:0.0919, r2:0.997, val_loss:0.1891, val_r2:0.814
e493, train_loss:0.0854, r2:0.997, val_loss:0.1922, val_r2:0.811
e494, train_loss:0.0961, r2:0.997, val_loss:0.1957, val_r2:0.807
e495, train_loss:0.0778, r2:0.997, val_loss:0.1974, val_r2:0.806
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e499, train_loss:0.0766, r2:0.997, val_loss:0.1974, val_r2:0.806
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e504, train_loss:0.0947, r2:0.997, val_loss:0.1954, val_r2:0.808
e505, train_loss:0.0928, r2:0.997, val_loss:0.1919, val_r2:0.811
e506, train_loss:0.0883, r2:0.997, val_loss:0.1899, val_r2:0.813
e507, train_loss:0.0927, r2:0.997, val_loss:0.1897, val_r2:0.813
e508, train_loss:0.0961, r2:0.997, val_loss:0.1916, val_r2:0.811

e509, train_loss:0.0902, r2:0.997, val_loss:0.1938, val_r2:0.809
e510, train_loss:0.0884, r2:0.997, val_loss:0.1973, val_r2:0.806
e511, train_loss:0.0862, r2:0.997, val_loss:0.1998, val_r2:0.803
e512, train_loss:0.0826, r2:0.997, val_loss:0.2005, val_r2:0.803
e513, train_loss:0.0929, r2:0.997, val_loss:0.2003, val_r2:0.803
e514, train_loss:0.0876, r2:0.997, val_loss:0.2005, val_r2:0.803
e515, train_loss:0.0877, r2:0.997, val_loss:0.1990, val_r2:0.804
e516, train_loss:0.0851, r2:0.997, val_loss:0.1963, val_r2:0.807
e517, train_loss:0.0893, r2:0.997, val_loss:0.1934, val_r2:0.809
e518, train_loss:0.0906, r2:0.997, val_loss:0.1893, val_r2:0.813
e519, train_loss:0.0818, r2:0.997, val_loss:0.1863, val_r2:0.816
e520, train_loss:0.0819, r2:0.997, val_loss:0.1846, val_r2:0.818
e521, train_loss:0.0819, r2:0.997, val_loss:0.1844, val_r2:0.818
e522, train_loss:0.0963, r2:0.997, val_loss:0.1888, val_r2:0.814
e523, train_loss:0.0874, r2:0.997, val_loss:0.1917, val_r2:0.811
e524, train_loss:0.0899, r2:0.997, val_loss:0.1936, val_r2:0.809
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e526, train_loss:0.0908, r2:0.996, val_loss:0.1954, val_r2:0.807
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e528, train_loss:0.0830, r2:0.997, val_loss:0.1963, val_r2:0.807
e529, train_loss:0.0932, r2:0.997, val_loss:0.1964, val_r2:0.807
e530, train_loss:0.0836, r2:0.997, val_loss:0.1963, val_r2:0.807
e531, train_loss:0.0872, r2:0.997, val_loss:0.1967, val_r2:0.806
e532, train_loss:0.0887, r2:0.997, val_loss:0.1959, val_r2:0.807
e533, train_loss:0.0924, r2:0.997, val_loss:0.1954, val_r2:0.808
e534, train_loss:0.0848, r2:0.997, val_loss:0.1959, val_r2:0.807
e535, train_loss:0.0876, r2:0.997, val_loss:0.1971, val_r2:0.806
e536, train_loss:0.0915, r2:0.997, val_loss:0.1962, val_r2:0.807
e537, train_loss:0.0910, r2:0.997, val_loss:0.1952, val_r2:0.808
e538, train_loss:0.0827, r2:0.997, val_loss:0.1947, val_r2:0.808
e539, train_loss:0.0774, r2:0.997, val_loss:0.1939, val_r2:0.809
e540, train_loss:0.0899, r2:0.997, val_loss:0.1938, val_r2:0.809
e541, train_loss:0.0899, r2:0.997, val_loss:0.1960, val_r2:0.807
e542, train_loss:0.0777, r2:0.997, val_loss:0.1972, val_r2:0.806
e543, train_loss:0.0966, r2:0.997, val_loss:0.1970, val_r2:0.806
e544, train_loss:0.0881, r2:0.997, val_loss:0.1965, val_r2:0.806
e545, train_loss:0.0870, r2:0.997, val_loss:0.1950, val_r2:0.808
e546, train_loss:0.0799, r2:0.997, val_loss:0.1930, val_r2:0.810
e547, train_loss:0.0839, r2:0.997, val_loss:0.1920, val_r2:0.811

e548, train_loss:0.0832, r2:0.997, val_loss:0.1918, val_r2:0.811
e549, train_loss:0.0765, r2:0.997, val_loss:0.1926, val_r2:0.810
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e551, train_loss:0.0809, r2:0.997, val_loss:0.1988, val_r2:0.804
e552, train_loss:0.0913, r2:0.997, val_loss:0.2011, val_r2:0.802
e553, train_loss:0.0945, r2:0.998, val_loss:0.2021, val_r2:0.801
e554, train_loss:0.0834, r2:0.997, val_loss:0.1999, val_r2:0.803
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e559, train_loss:0.0849, r2:0.997, val_loss:0.1971, val_r2:0.806
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e561, train_loss:0.0902, r2:0.998, val_loss:0.1965, val_r2:0.806
e562, train_loss:0.0879, r2:0.997, val_loss:0.1959, val_r2:0.807
e563, train_loss:0.0934, r2:0.997, val_loss:0.1953, val_r2:0.808
e564, train_loss:0.0907, r2:0.997, val_loss:0.1965, val_r2:0.806
e565, train_loss:0.0916, r2:0.997, val_loss:0.1950, val_r2:0.808
e566, train_loss:0.0938, r2:0.997, val_loss:0.1928, val_r2:0.810
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e568, train_loss:0.0937, r2:0.997, val_loss:0.1885, val_r2:0.814
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e570, train_loss:0.0901, r2:0.997, val_loss:0.1910, val_r2:0.812
e571, train_loss:0.0883, r2:0.997, val_loss:0.1941, val_r2:0.809
e572, train_loss:0.0957, r2:0.997, val_loss:0.1967, val_r2:0.806
e573, train_loss:0.0877, r2:0.997, val_loss:0.2001, val_r2:0.803
e574, train_loss:0.0948, r2:0.996, val_loss:0.2010, val_r2:0.802
e575, train_loss:0.0913, r2:0.997, val_loss:0.2010, val_r2:0.802
e576, train_loss:0.0823, r2:0.997, val_loss:0.2009, val_r2:0.802
e577, train_loss:0.0877, r2:0.997, val_loss:0.1990, val_r2:0.804
e578, train_loss:0.0853, r2:0.997, val_loss:0.1962, val_r2:0.807
e579, train_loss:0.0899, r2:0.998, val_loss:0.1950, val_r2:0.808
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e581, train_loss:0.0832, r2:0.997, val_loss:0.1933, val_r2:0.810
e582, train_loss:0.0856, r2:0.997, val_loss:0.1932, val_r2:0.810
e583, train_loss:0.0884, r2:0.997, val_loss:0.1913, val_r2:0.812
e584, train_loss:0.0881, r2:0.998, val_loss:0.1921, val_r2:0.811
e585, train_loss:0.0842, r2:0.997, val_loss:0.1938, val_r2:0.809
e586, train_loss:0.0870, r2:0.997, val_loss:0.1947, val_r2:0.808

e587, train_loss:0.0953, r2:0.997, val_loss:0.1955, val_r2:0.807
e588, train_loss:0.0930, r2:0.997, val_loss:0.1981, val_r2:0.805
e589, train_loss:0.0803, r2:0.997, val_loss:0.1983, val_r2:0.805
e590, train_loss:0.0920, r2:0.997, val_loss:0.1973, val_r2:0.806
e591, train_loss:0.0877, r2:0.997, val_loss:0.1949, val_r2:0.808
e592, train_loss:0.0819, r2:0.997, val_loss:0.1914, val_r2:0.811
e593, train_loss:0.0858, r2:0.997, val_loss:0.1890, val_r2:0.814
e594, train_loss:0.0956, r2:0.997, val_loss:0.1914, val_r2:0.811
e595, train_loss:0.0928, r2:0.996, val_loss:0.1945, val_r2:0.808
e596, train_loss:0.0939, r2:0.996, val_loss:0.1943, val_r2:0.809
e597, train_loss:0.0976, r2:0.997, val_loss:0.1947, val_r2:0.808
e598, train_loss:0.0827, r2:0.997, val_loss:0.1942, val_r2:0.809
e599, train_loss:0.0911, r2:0.997, val_loss:0.1925, val_r2:0.810
e600, train_loss:0.0804, r2:0.998, val_loss:0.1927, val_r2:0.810
e601, train_loss:0.0841, r2:0.997, val_loss:0.1949, val_r2:0.808
e602, train_loss:0.0902, r2:0.997, val_loss:0.1994, val_r2:0.804
e603, train_loss:0.0920, r2:0.997, val_loss:0.2027, val_r2:0.800
e604, train_loss:0.0816, r2:0.998, val_loss:0.2046, val_r2:0.798
e605, train_loss:0.0928, r2:0.997, val_loss:0.2044, val_r2:0.799
e606, train_loss:0.0951, r2:0.997, val_loss:0.2015, val_r2:0.802
e607, train_loss:0.0925, r2:0.998, val_loss:0.1997, val_r2:0.803
e608, train_loss:0.0883, r2:0.997, val_loss:0.1998, val_r2:0.803
e609, train_loss:0.0866, r2:0.996, val_loss:0.1987, val_r2:0.804
e610, train_loss:0.0893, r2:0.997, val_loss:0.1974, val_r2:0.806
e611, train_loss:0.0860, r2:0.997, val_loss:0.1979, val_r2:0.805
e612, train_loss:0.0972, r2:0.996, val_loss:0.1959, val_r2:0.807
e613, train_loss:0.0892, r2:0.997, val_loss:0.1954, val_r2:0.808
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e615, train_loss:0.0852, r2:0.996, val_loss:0.1978, val_r2:0.805
e616, train_loss:0.0820, r2:0.997, val_loss:0.1989, val_r2:0.804
e617, train_loss:0.0827, r2:0.998, val_loss:0.2018, val_r2:0.801
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e622, train_loss:0.0841, r2:0.997, val_loss:0.1969, val_r2:0.806
e623, train_loss:0.0984, r2:0.997, val_loss:0.1964, val_r2:0.807
e624, train_loss:0.0825, r2:0.997, val_loss:0.1971, val_r2:0.806
e625, train_loss:0.0781, r2:0.998, val_loss:0.1984, val_r2:0.805

e626, train_loss:0.0941, r2:0.997, val_loss:0.2000, val_r2:0.803
e627, train_loss:0.0905, r2:0.997, val_loss:0.2001, val_r2:0.803
e628, train_loss:0.0823, r2:0.997, val_loss:0.1998, val_r2:0.803
e629, train_loss:0.0782, r2:0.997, val_loss:0.1996, val_r2:0.803
e630, train_loss:0.0965, r2:0.997, val_loss:0.2008, val_r2:0.802
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e632, train_loss:0.0892, r2:0.997, val_loss:0.2055, val_r2:0.798
e633, train_loss:0.0789, r2:0.997, val_loss:0.2042, val_r2:0.799
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e636, train_loss:0.0960, r2:0.997, val_loss:0.1937, val_r2:0.809
e637, train_loss:0.0740, r2:0.998, val_loss:0.1915, val_r2:0.811
e638, train_loss:0.0898, r2:0.997, val_loss:0.1914, val_r2:0.811
e639, train_loss:0.0822, r2:0.997, val_loss:0.1923, val_r2:0.811
e640, train_loss:0.0880, r2:0.997, val_loss:0.1962, val_r2:0.807
e641, train_loss:0.0890, r2:0.997, val_loss:0.2007, val_r2:0.802
e642, train_loss:0.0878, r2:0.997, val_loss:0.2046, val_r2:0.798
e643, train_loss:0.0844, r2:0.998, val_loss:0.2052, val_r2:0.798
e644, train_loss:0.0971, r2:0.998, val_loss:0.2041, val_r2:0.799
e645, train_loss:0.0922, r2:0.997, val_loss:0.2014, val_r2:0.802
e646, train_loss:0.0903, r2:0.998, val_loss:0.1981, val_r2:0.805
e647, train_loss:0.0895, r2:0.997, val_loss:0.1964, val_r2:0.807
e648, train_loss:0.0854, r2:0.997, val_loss:0.1967, val_r2:0.806
e649, train_loss:0.0842, r2:0.998, val_loss:0.1990, val_r2:0.804
e650, train_loss:0.0821, r2:0.998, val_loss:0.2015, val_r2:0.801
e651, train_loss:0.0818, r2:0.998, val_loss:0.2045, val_r2:0.799
e652, train_loss:0.0772, r2:0.998, val_loss:0.2057, val_r2:0.797
e653, train_loss:0.0809, r2:0.998, val_loss:0.2052, val_r2:0.798
e654, train_loss:0.0784, r2:0.998, val_loss:0.2037, val_r2:0.799
e655, train_loss:0.0800, r2:0.998, val_loss:0.2012, val_r2:0.802
e656, train_loss:0.0849, r2:0.998, val_loss:0.2005, val_r2:0.802
e657, train_loss:0.0860, r2:0.997, val_loss:0.2011, val_r2:0.802
e658, train_loss:0.0863, r2:0.997, val_loss:0.2033, val_r2:0.800
e659, train_loss:0.0965, r2:0.998, val_loss:0.2030, val_r2:0.800
e660, train_loss:0.0916, r2:0.997, val_loss:0.2023, val_r2:0.801
e661, train_loss:0.0854, r2:0.997, val_loss:0.2005, val_r2:0.802
e662, train_loss:0.0849, r2:0.997, val_loss:0.1984, val_r2:0.805
e663, train_loss:0.0885, r2:0.998, val_loss:0.1981, val_r2:0.805
e664, train_loss:0.0882, r2:0.997, val_loss:0.1990, val_r2:0.804

e665, train_loss:0.0968, r2:0.997, val_loss:0.2026, val_r2:0.800
e666, train_loss:0.0973, r2:0.998, val_loss:0.2067, val_r2:0.796
e667, train_loss:0.0758, r2:0.998, val_loss:0.2080, val_r2:0.795
e668, train_loss:0.0959, r2:0.997, val_loss:0.2065, val_r2:0.797
e669, train_loss:0.0878, r2:0.997, val_loss:0.2030, val_r2:0.800
e670, train_loss:0.0861, r2:0.997, val_loss:0.1990, val_r2:0.804
e671, train_loss:0.0955, r2:0.998, val_loss:0.1969, val_r2:0.806
e672, train_loss:0.0805, r2:0.997, val_loss:0.1944, val_r2:0.809
e673, train_loss:0.0975, r2:0.997, val_loss:0.1939, val_r2:0.809
e674, train_loss:0.0828, r2:0.998, val_loss:0.1953, val_r2:0.808
e675, train_loss:0.0843, r2:0.997, val_loss:0.1981, val_r2:0.805
e676, train_loss:0.0891, r2:0.997, val_loss:0.2018, val_r2:0.801
e677, train_loss:0.0795, r2:0.998, val_loss:0.2060, val_r2:0.797
e678, train_loss:0.0847, r2:0.997, val_loss:0.2060, val_r2:0.797
e679, train_loss:0.0797, r2:0.998, val_loss:0.2053, val_r2:0.798
e680, train_loss:0.0848, r2:0.998, val_loss:0.2061, val_r2:0.797
e681, train_loss:0.0831, r2:0.997, val_loss:0.2063, val_r2:0.797
e682, train_loss:0.0979, r2:0.997, val_loss:0.2068, val_r2:0.796
e683, train_loss:0.0908, r2:0.998, val_loss:0.2075, val_r2:0.796
e684, train_loss:0.0894, r2:0.997, val_loss:0.2053, val_r2:0.798
e685, train_loss:0.0845, r2:0.998, val_loss:0.2015, val_r2:0.802
e686, train_loss:0.0924, r2:0.998, val_loss:0.1962, val_r2:0.807
e687, train_loss:0.0783, r2:0.998, val_loss:0.1919, val_r2:0.811
e688, train_loss:0.0981, r2:0.998, val_loss:0.1905, val_r2:0.812
e689, train_loss:0.0776, r2:0.997, val_loss:0.1913, val_r2:0.812
e690, train_loss:0.0883, r2:0.997, val_loss:0.1934, val_r2:0.809
e691, train_loss:0.0879, r2:0.998, val_loss:0.1980, val_r2:0.805
e692, train_loss:0.0949, r2:0.998, val_loss:0.2023, val_r2:0.801
e693, train_loss:0.0841, r2:0.997, val_loss:0.2045, val_r2:0.799
e694, train_loss:0.0939, r2:0.997, val_loss:0.2056, val_r2:0.797
e695, train_loss:0.0926, r2:0.998, val_loss:0.2051, val_r2:0.798
e696, train_loss:0.0830, r2:0.997, val_loss:0.2021, val_r2:0.801
e697, train_loss:0.0821, r2:0.998, val_loss:0.1993, val_r2:0.804
e698, train_loss:0.0878, r2:0.998, val_loss:0.1976, val_r2:0.805
e699, train_loss:0.0838, r2:0.997, val_loss:0.1970, val_r2:0.806
e700, train_loss:0.0858, r2:0.997, val_loss:0.1971, val_r2:0.806
e701, train_loss:0.0922, r2:0.997, val_loss:0.1965, val_r2:0.806
e702, train_loss:0.0835, r2:0.997, val_loss:0.1947, val_r2:0.808
e703, train_loss:0.0958, r2:0.998, val_loss:0.1945, val_r2:0.808

e704, train_loss:0.0865, r2:0.997, val_loss:0.1947, val_r2:0.808
e705, train_loss:0.0779, r2:0.998, val_loss:0.1961, val_r2:0.807
e706, train_loss:0.0908, r2:0.998, val_loss:0.2007, val_r2:0.802
e707, train_loss:0.0887, r2:0.997, val_loss:0.2025, val_r2:0.801
e708, train_loss:0.0878, r2:0.998, val_loss:0.2028, val_r2:0.800
e709, train_loss:0.0827, r2:0.997, val_loss:0.2033, val_r2:0.800
e710, train_loss:0.0867, r2:0.997, val_loss:0.2012, val_r2:0.802
e711, train_loss:0.0824, r2:0.997, val_loss:0.1983, val_r2:0.805
e712, train_loss:0.0821, r2:0.998, val_loss:0.1956, val_r2:0.807
e713, train_loss:0.0872, r2:0.997, val_loss:0.1946, val_r2:0.808
e714, train_loss:0.0964, r2:0.998, val_loss:0.1958, val_r2:0.807
e715, train_loss:0.0972, r2:0.998, val_loss:0.1976, val_r2:0.805
e716, train_loss:0.0913, r2:0.998, val_loss:0.1990, val_r2:0.804
e717, train_loss:0.0821, r2:0.998, val_loss:0.1992, val_r2:0.804
e718, train_loss:0.0913, r2:0.998, val_loss:0.1997, val_r2:0.803
e719, train_loss:0.0852, r2:0.998, val_loss:0.2011, val_r2:0.802
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e721, train_loss:0.0835, r2:0.998, val_loss:0.2008, val_r2:0.802
e722, train_loss:0.0890, r2:0.998, val_loss:0.2014, val_r2:0.802
e723, train_loss:0.0976, r2:0.997, val_loss:0.2003, val_r2:0.803
e724, train_loss:0.0838, r2:0.998, val_loss:0.1992, val_r2:0.804
e725, train_loss:0.0797, r2:0.998, val_loss:0.1984, val_r2:0.805
e726, train_loss:0.0878, r2:0.997, val_loss:0.1960, val_r2:0.807
e727, train_loss:0.0905, r2:0.997, val_loss:0.1938, val_r2:0.809
e728, train_loss:0.0908, r2:0.998, val_loss:0.1918, val_r2:0.811
e729, train_loss:0.0748, r2:0.998, val_loss:0.1895, val_r2:0.813
e730, train_loss:0.0902, r2:0.998, val_loss:0.1881, val_r2:0.815
e731, train_loss:0.0881, r2:0.998, val_loss:0.1889, val_r2:0.814
e732, train_loss:0.0857, r2:0.998, val_loss:0.1925, val_r2:0.810
e733, train_loss:0.0852, r2:0.997, val_loss:0.1957, val_r2:0.807
e734, train_loss:0.0872, r2:0.998, val_loss:0.1989, val_r2:0.804
e735, train_loss:0.0924, r2:0.998, val_loss:0.2023, val_r2:0.801
e736, train_loss:0.0831, r2:0.997, val_loss:0.2038, val_r2:0.799
e737, train_loss:0.0858, r2:0.998, val_loss:0.2040, val_r2:0.799
e738, train_loss:0.0886, r2:0.998, val_loss:0.2023, val_r2:0.801
e739, train_loss:0.0955, r2:0.998, val_loss:0.1989, val_r2:0.804
e740, train_loss:0.0841, r2:0.998, val_loss:0.1954, val_r2:0.808
e741, train_loss:0.0911, r2:0.998, val_loss:0.1941, val_r2:0.809
e742, train_loss:0.0856, r2:0.998, val_loss:0.1948, val_r2:0.808

e743, train_loss:0.0859, r2:0.997, val_loss:0.1952, val_r2:0.808
e744, train_loss:0.0804, r2:0.998, val_loss:0.1941, val_r2:0.809
e745, train_loss:0.0838, r2:0.998, val_loss:0.1942, val_r2:0.809
e746, train_loss:0.0867, r2:0.997, val_loss:0.1940, val_r2:0.809
e747, train_loss:0.0873, r2:0.998, val_loss:0.1957, val_r2:0.807
e748, train_loss:0.0870, r2:0.998, val_loss:0.1985, val_r2:0.804
e749, train_loss:0.0901, r2:0.997, val_loss:0.2012, val_r2:0.802
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e751, train_loss:0.0955, r2:0.998, val_loss:0.2001, val_r2:0.803
e752, train_loss:0.0859, r2:0.998, val_loss:0.1975, val_r2:0.805
e753, train_loss:0.0855, r2:0.998, val_loss:0.1934, val_r2:0.810
e754, train_loss:0.0910, r2:0.998, val_loss:0.1906, val_r2:0.812
e755, train_loss:0.0926, r2:0.998, val_loss:0.1901, val_r2:0.813
e756, train_loss:0.0893, r2:0.997, val_loss:0.1911, val_r2:0.812
e757, train_loss:0.0779, r2:0.998, val_loss:0.1942, val_r2:0.809
e758, train_loss:0.0896, r2:0.998, val_loss:0.1977, val_r2:0.805
e759, train_loss:0.0921, r2:0.998, val_loss:0.2014, val_r2:0.802
e760, train_loss:0.0783, r2:0.998, val_loss:0.2039, val_r2:0.799
e761, train_loss:0.0813, r2:0.998, val_loss:0.2046, val_r2:0.798
e762, train_loss:0.0812, r2:0.998, val_loss:0.2044, val_r2:0.799
e763, train_loss:0.0885, r2:0.998, val_loss:0.2034, val_r2:0.800
e764, train_loss:0.0846, r2:0.998, val_loss:0.2029, val_r2:0.800
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e767, train_loss:0.0890, r2:0.998, val_loss:0.1975, val_r2:0.805
e768, train_loss:0.0915, r2:0.998, val_loss:0.1953, val_r2:0.808
e769, train_loss:0.0841, r2:0.998, val_loss:0.1931, val_r2:0.810
e770, train_loss:0.0820, r2:0.998, val_loss:0.1929, val_r2:0.810
e771, train_loss:0.0887, r2:0.998, val_loss:0.1940, val_r2:0.809
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e773, train_loss:0.0915, r2:0.998, val_loss:0.1944, val_r2:0.808
e774, train_loss:0.0856, r2:0.998, val_loss:0.1954, val_r2:0.807
e775, train_loss:0.0889, r2:0.997, val_loss:0.1958, val_r2:0.807
e776, train_loss:0.0877, r2:0.998, val_loss:0.1966, val_r2:0.806
e777, train_loss:0.0926, r2:0.998, val_loss:0.1975, val_r2:0.805
e778, train_loss:0.0770, r2:0.997, val_loss:0.1965, val_r2:0.806
e779, train_loss:0.0901, r2:0.998, val_loss:0.1968, val_r2:0.806
e780, train_loss:0.0775, r2:0.998, val_loss:0.1964, val_r2:0.807
e781, train_loss:0.0865, r2:0.998, val_loss:0.1965, val_r2:0.806

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e784, train_loss:0.0911, r2:0.998, val_loss:0.1983, val_r2:0.805
e785, train_loss:0.0919, r2:0.998, val_loss:0.2000, val_r2:0.803
e786, train_loss:0.0826, r2:0.998, val_loss:0.2015, val_r2:0.802
e787, train_loss:0.0920, r2:0.998, val_loss:0.2027, val_r2:0.800
e788, train_loss:0.0844, r2:0.998, val_loss:0.2020, val_r2:0.801
e789, train_loss:0.0846, r2:0.998, val_loss:0.1995, val_r2:0.804
e790, train_loss:0.0815, r2:0.998, val_loss:0.1960, val_r2:0.807
e791, train_loss:0.0955, r2:0.998, val_loss:0.1941, val_r2:0.809
e792, train_loss:0.0810, r2:0.998, val_loss:0.1933, val_r2:0.810
e793, train_loss:0.0807, r2:0.998, val_loss:0.1937, val_r2:0.809
e794, train_loss:0.0772, r2:0.998, val_loss:0.1947, val_r2:0.808
e795, train_loss:0.0901, r2:0.998, val_loss:0.1953, val_r2:0.808
e796, train_loss:0.0812, r2:0.998, val_loss:0.1952, val_r2:0.808
e797, train_loss:0.0812, r2:0.998, val_loss:0.1958, val_r2:0.807
e798, train_loss:0.0884, r2:0.998, val_loss:0.1978, val_r2:0.805
e799, train_loss:0.0926, r2:0.998, val_loss:0.2010, val_r2:0.802
e800, train_loss:0.0865, r2:0.998, val_loss:0.2012, val_r2:0.802
e801, train_loss:0.0830, r2:0.998, val_loss:0.2005, val_r2:0.802
e802, train_loss:0.0820, r2:0.997, val_loss:0.1988, val_r2:0.804
e803, train_loss:0.0771, r2:0.998, val_loss:0.1963, val_r2:0.807
e804, train_loss:0.0817, r2:0.998, val_loss:0.1949, val_r2:0.808
e805, train_loss:0.0883, r2:0.998, val_loss:0.1944, val_r2:0.808
e806, train_loss:0.0845, r2:0.998, val_loss:0.1947, val_r2:0.808
e807, train_loss:0.0904, r2:0.998, val_loss:0.1969, val_r2:0.806
e808, train_loss:0.0746, r2:0.998, val_loss:0.1979, val_r2:0.805
e809, train_loss:0.0817, r2:0.998, val_loss:0.1967, val_r2:0.806
e810, train_loss:0.0864, r2:0.998, val_loss:0.1959, val_r2:0.807
e811, train_loss:0.0780, r2:0.998, val_loss:0.1944, val_r2:0.808
e812, train_loss:0.0771, r2:0.998, val_loss:0.1927, val_r2:0.810
e813, train_loss:0.0787, r2:0.998, val_loss:0.1934, val_r2:0.809
e814, train_loss:0.0909, r2:0.998, val_loss:0.1950, val_r2:0.808
e815, train_loss:0.0838, r2:0.998, val_loss:0.1954, val_r2:0.808
e816, train_loss:0.0900, r2:0.998, val_loss:0.1969, val_r2:0.806
e817, train_loss:0.0849, r2:0.998, val_loss:0.1984, val_r2:0.805
e818, train_loss:0.0856, r2:0.998, val_loss:0.1978, val_r2:0.805
e819, train_loss:0.0825, r2:0.998, val_loss:0.1979, val_r2:0.805
e820, train_loss:0.0862, r2:0.998, val_loss:0.1990, val_r2:0.804

e821, train_loss:0.0836, r2:0.998, val_loss:0.2004, val_r2:0.803
e822, train_loss:0.0882, r2:0.998, val_loss:0.2009, val_r2:0.802
e823, train_loss:0.0891, r2:0.998, val_loss:0.2004, val_r2:0.803
e824, train_loss:0.0731, r2:0.998, val_loss:0.1982, val_r2:0.805
e825, train_loss:0.0826, r2:0.998, val_loss:0.1949, val_r2:0.808
e826, train_loss:0.0876, r2:0.998, val_loss:0.1923, val_r2:0.811
e827, train_loss:0.0936, r2:0.998, val_loss:0.1946, val_r2:0.808
e828, train_loss:0.0828, r2:0.997, val_loss:0.1962, val_r2:0.807
e829, train_loss:0.0911, r2:0.997, val_loss:0.1954, val_r2:0.807
e830, train_loss:0.1006, r2:0.998, val_loss:0.1968, val_r2:0.806
e831, train_loss:0.0857, r2:0.997, val_loss:0.1966, val_r2:0.806
e832, train_loss:0.0861, r2:0.998, val_loss:0.1957, val_r2:0.807
e833, train_loss:0.0817, r2:0.998, val_loss:0.1959, val_r2:0.807
e834, train_loss:0.0850, r2:0.997, val_loss:0.1975, val_r2:0.805
e835, train_loss:0.0800, r2:0.997, val_loss:0.2000, val_r2:0.803
e836, train_loss:0.0859, r2:0.998, val_loss:0.2023, val_r2:0.801
e837, train_loss:0.0823, r2:0.997, val_loss:0.2023, val_r2:0.801
e838, train_loss:0.0844, r2:0.998, val_loss:0.2005, val_r2:0.803
e839, train_loss:0.0916, r2:0.998, val_loss:0.1985, val_r2:0.804
e840, train_loss:0.0837, r2:0.998, val_loss:0.1963, val_r2:0.807
e841, train_loss:0.0802, r2:0.998, val_loss:0.1942, val_r2:0.809
e842, train_loss:0.0799, r2:0.998, val_loss:0.1929, val_r2:0.810
e843, train_loss:0.0779, r2:0.998, val_loss:0.1929, val_r2:0.810
e844, train_loss:0.0948, r2:0.998, val_loss:0.1938, val_r2:0.809
e845, train_loss:0.0800, r2:0.998, val_loss:0.1969, val_r2:0.806
e846, train_loss:0.0786, r2:0.998, val_loss:0.1997, val_r2:0.803
e847, train_loss:0.0894, r2:0.998, val_loss:0.2028, val_r2:0.800
e848, train_loss:0.0794, r2:0.998, val_loss:0.2045, val_r2:0.799
e849, train_loss:0.0796, r2:0.998, val_loss:0.2043, val_r2:0.799
e850, train_loss:0.0900, r2:0.997, val_loss:0.2028, val_r2:0.800
e851, train_loss:0.0840, r2:0.998, val_loss:0.2032, val_r2:0.800
e852, train_loss:0.0824, r2:0.997, val_loss:0.2040, val_r2:0.799
e853, train_loss:0.0870, r2:0.997, val_loss:0.2026, val_r2:0.800
e854, train_loss:0.0772, r2:0.998, val_loss:0.1997, val_r2:0.803
e855, train_loss:0.0819, r2:0.998, val_loss:0.1971, val_r2:0.806
e856, train_loss:0.0836, r2:0.997, val_loss:0.1947, val_r2:0.808
e857, train_loss:0.0885, r2:0.998, val_loss:0.1955, val_r2:0.807
e858, train_loss:0.0809, r2:0.998, val_loss:0.1950, val_r2:0.808
e859, train_loss:0.0844, r2:0.997, val_loss:0.1946, val_r2:0.808

e860, train_loss:0.0835, r2:0.998, val_loss:0.1957, val_r2:0.807
e861, train_loss:0.0910, r2:0.998, val_loss:0.1960, val_r2:0.807
e862, train_loss:0.0859, r2:0.998, val_loss:0.1941, val_r2:0.809
e863, train_loss:0.0775, r2:0.998, val_loss:0.1916, val_r2:0.811
e864, train_loss:0.0853, r2:0.998, val_loss:0.1905, val_r2:0.812
e865, train_loss:0.0895, r2:0.998, val_loss:0.1917, val_r2:0.811
e866, train_loss:0.0928, r2:0.998, val_loss:0.1954, val_r2:0.807
e867, train_loss:0.0960, r2:0.998, val_loss:0.1985, val_r2:0.804
e868, train_loss:0.0819, r2:0.998, val_loss:0.1987, val_r2:0.804
e869, train_loss:0.0803, r2:0.998, val_loss:0.1976, val_r2:0.805
e870, train_loss:0.0985, r2:0.997, val_loss:0.1946, val_r2:0.808
e871, train_loss:0.0941, r2:0.998, val_loss:0.1938, val_r2:0.809
e872, train_loss:0.0885, r2:0.998, val_loss:0.1955, val_r2:0.807
e873, train_loss:0.0813, r2:0.997, val_loss:0.1984, val_r2:0.805
e874, train_loss:0.0887, r2:0.998, val_loss:0.2024, val_r2:0.801
e875, train_loss:0.0766, r2:0.998, val_loss:0.2034, val_r2:0.800
e876, train_loss:0.0887, r2:0.997, val_loss:0.1999, val_r2:0.803
e877, train_loss:0.0881, r2:0.998, val_loss:0.1950, val_r2:0.808
e878, train_loss:0.0840, r2:0.998, val_loss:0.1907, val_r2:0.812
e879, train_loss:0.0851, r2:0.998, val_loss:0.1891, val_r2:0.814
e880, train_loss:0.0832, r2:0.997, val_loss:0.1914, val_r2:0.811
e881, train_loss:0.0831, r2:0.998, val_loss:0.1942, val_r2:0.809
e882, train_loss:0.0909, r2:0.998, val_loss:0.1958, val_r2:0.807
e883, train_loss:0.0912, r2:0.997, val_loss:0.1964, val_r2:0.807
e884, train_loss:0.0886, r2:0.998, val_loss:0.1948, val_r2:0.808
e885, train_loss:0.0811, r2:0.998, val_loss:0.1937, val_r2:0.809
e886, train_loss:0.0870, r2:0.998, val_loss:0.1936, val_r2:0.809
e887, train_loss:0.0796, r2:0.998, val_loss:0.1955, val_r2:0.807
e888, train_loss:0.0792, r2:0.998, val_loss:0.1986, val_r2:0.804
e889, train_loss:0.0804, r2:0.998, val_loss:0.2005, val_r2:0.802
e890, train_loss:0.0863, r2:0.998, val_loss:0.2010, val_r2:0.802
e891, train_loss:0.0886, r2:0.998, val_loss:0.1981, val_r2:0.805
e892, train_loss:0.0881, r2:0.998, val_loss:0.1939, val_r2:0.809
e893, train_loss:0.0915, r2:0.998, val_loss:0.1897, val_r2:0.813
e894, train_loss:0.0907, r2:0.998, val_loss:0.1872, val_r2:0.816
e895, train_loss:0.0824, r2:0.998, val_loss:0.1883, val_r2:0.814
e896, train_loss:0.0870, r2:0.998, val_loss:0.1907, val_r2:0.812
e897, train_loss:0.0875, r2:0.998, val_loss:0.1927, val_r2:0.810
e898, train_loss:0.0866, r2:0.998, val_loss:0.1942, val_r2:0.809

e899, train_loss:0.0841, r2:0.997, val_loss:0.1940, val_r2:0.809
e900, train_loss:0.0834, r2:0.998, val_loss:0.1933, val_r2:0.810
e901, train_loss:0.0803, r2:0.998, val_loss:0.1950, val_r2:0.808
e902, train_loss:0.0828, r2:0.997, val_loss:0.1970, val_r2:0.806
e903, train_loss:0.0923, r2:0.997, val_loss:0.1997, val_r2:0.803
e904, train_loss:0.0916, r2:0.998, val_loss:0.2014, val_r2:0.802
e905, train_loss:0.0912, r2:0.997, val_loss:0.2007, val_r2:0.802
e906, train_loss:0.0859, r2:0.997, val_loss:0.1963, val_r2:0.807
e907, train_loss:0.0814, r2:0.997, val_loss:0.1897, val_r2:0.813
e908, train_loss:0.0837, r2:0.998, val_loss:0.1848, val_r2:0.818
e909, train_loss:0.0849, r2:0.997, val_loss:0.1839, val_r2:0.819
e910, train_loss:0.0918, r2:0.996, val_loss:0.1831, val_r2:0.820
e911, train_loss:0.0862, r2:0.997, val_loss:0.1846, val_r2:0.818
e912, train_loss:0.0894, r2:0.997, val_loss:0.1863, val_r2:0.816
e913, train_loss:0.0910, r2:0.997, val_loss:0.1879, val_r2:0.815
e914, train_loss:0.0858, r2:0.998, val_loss:0.1896, val_r2:0.813
e915, train_loss:0.0875, r2:0.998, val_loss:0.1909, val_r2:0.812
e916, train_loss:0.0814, r2:0.998, val_loss:0.1938, val_r2:0.809
e917, train_loss:0.0863, r2:0.998, val_loss:0.1977, val_r2:0.805
e918, train_loss:0.0885, r2:0.998, val_loss:0.2014, val_r2:0.802
e919, train_loss:0.0920, r2:0.998, val_loss:0.2033, val_r2:0.800
e920, train_loss:0.0874, r2:0.998, val_loss:0.2005, val_r2:0.802
e921, train_loss:0.0890, r2:0.998, val_loss:0.1963, val_r2:0.807
e922, train_loss:0.0951, r2:0.997, val_loss:0.1923, val_r2:0.811
e923, train_loss:0.0822, r2:0.998, val_loss:0.1903, val_r2:0.813
e924, train_loss:0.0789, r2:0.998, val_loss:0.1894, val_r2:0.813
e925, train_loss:0.0823, r2:0.998, val_loss:0.1895, val_r2:0.813
e926, train_loss:0.0872, r2:0.998, val_loss:0.1906, val_r2:0.812
e927, train_loss:0.0860, r2:0.998, val_loss:0.1922, val_r2:0.811
e928, train_loss:0.0830, r2:0.998, val_loss:0.1928, val_r2:0.810
e929, train_loss:0.0857, r2:0.998, val_loss:0.1927, val_r2:0.810
e930, train_loss:0.0918, r2:0.998, val_loss:0.1939, val_r2:0.809
e931, train_loss:0.0848, r2:0.997, val_loss:0.1945, val_r2:0.808
e932, train_loss:0.0861, r2:0.997, val_loss:0.1930, val_r2:0.810
e933, train_loss:0.0849, r2:0.998, val_loss:0.1906, val_r2:0.812
e934, train_loss:0.0927, r2:0.997, val_loss:0.1882, val_r2:0.815
e935, train_loss:0.0739, r2:0.997, val_loss:0.1849, val_r2:0.818
e936, train_loss:0.0849, r2:0.998, val_loss:0.1833, val_r2:0.819
e937, train_loss:0.0809, r2:0.998, val_loss:0.1854, val_r2:0.817

e938, train_loss:0.0942, r2:0.997, val_loss:0.1881, val_r2:0.815
e939, train_loss:0.0743, r2:0.998, val_loss:0.1904, val_r2:0.812
e940, train_loss:0.0757, r2:0.997, val_loss:0.1937, val_r2:0.809
e941, train_loss:0.0928, r2:0.996, val_loss:0.1920, val_r2:0.811
e942, train_loss:0.0860, r2:0.998, val_loss:0.1937, val_r2:0.809
e943, train_loss:0.0802, r2:0.997, val_loss:0.1949, val_r2:0.808
e944, train_loss:0.0811, r2:0.996, val_loss:0.1906, val_r2:0.812
e945, train_loss:0.0938, r2:0.998, val_loss:0.1898, val_r2:0.813
e946, train_loss:0.0857, r2:0.997, val_loss:0.1893, val_r2:0.813
e947, train_loss:0.0931, r2:0.996, val_loss:0.1855, val_r2:0.817
e948, train_loss:0.0866, r2:0.998, val_loss:0.1875, val_r2:0.815
e949, train_loss:0.0785, r2:0.997, val_loss:0.1891, val_r2:0.814
e950, train_loss:0.0824, r2:0.997, val_loss:0.1906, val_r2:0.812
e951, train_loss:0.0862, r2:0.998, val_loss:0.1936, val_r2:0.809
e952, train_loss:0.0774, r2:0.998, val_loss:0.1959, val_r2:0.807
e953, train_loss:0.0753, r2:0.997, val_loss:0.1957, val_r2:0.807
e954, train_loss:0.0886, r2:0.998, val_loss:0.1943, val_r2:0.809
e955, train_loss:0.0920, r2:0.998, val_loss:0.1921, val_r2:0.811
e956, train_loss:0.0877, r2:0.997, val_loss:0.1898, val_r2:0.813
e957, train_loss:0.0827, r2:0.997, val_loss:0.1881, val_r2:0.815
e958, train_loss:0.0989, r2:0.998, val_loss:0.1872, val_r2:0.816
e959, train_loss:0.0847, r2:0.998, val_loss:0.1862, val_r2:0.817
e960, train_loss:0.0783, r2:0.998, val_loss:0.1858, val_r2:0.817
e961, train_loss:0.0859, r2:0.997, val_loss:0.1858, val_r2:0.817
e962, train_loss:0.0879, r2:0.998, val_loss:0.1875, val_r2:0.815
e963, train_loss:0.0907, r2:0.998, val_loss:0.1897, val_r2:0.813
e964, train_loss:0.0872, r2:0.998, val_loss:0.1916, val_r2:0.811
e965, train_loss:0.0872, r2:0.998, val_loss:0.1936, val_r2:0.809
e966, train_loss:0.0772, r2:0.998, val_loss:0.1935, val_r2:0.809
e967, train_loss:0.0883, r2:0.998, val_loss:0.1918, val_r2:0.811
e968, train_loss:0.0714, r2:0.998, val_loss:0.1883, val_r2:0.814
e969, train_loss:0.0761, r2:0.998, val_loss:0.1852, val_r2:0.818
e970, train_loss:0.0801, r2:0.998, val_loss:0.1838, val_r2:0.819
e971, train_loss:0.0843, r2:0.998, val_loss:0.1847, val_r2:0.818
e972, train_loss:0.0863, r2:0.997, val_loss:0.1877, val_r2:0.815
e973, train_loss:0.0845, r2:0.997, val_loss:0.1926, val_r2:0.810
e974, train_loss:0.0798, r2:0.998, val_loss:0.1946, val_r2:0.808
e975, train_loss:0.0843, r2:0.998, val_loss:0.1966, val_r2:0.806
e976, train_loss:0.0864, r2:0.996, val_loss:0.1953, val_r2:0.808

e977, train_loss:0.0780, r2:0.997, val_loss:0.1908, val_r2:0.812
e978, train_loss:0.0900, r2:0.997, val_loss:0.1885, val_r2:0.814
e979, train_loss:0.0732, r2:0.996, val_loss:0.1870, val_r2:0.816
e980, train_loss:0.0817, r2:0.997, val_loss:0.1866, val_r2:0.816
e981, train_loss:0.0803, r2:0.998, val_loss:0.1865, val_r2:0.816
e982, train_loss:0.0788, r2:0.997, val_loss:0.1843, val_r2:0.818
e983, train_loss:0.0891, r2:0.997, val_loss:0.1805, val_r2:0.822
e984, train_loss:0.0788, r2:0.997, val_loss:0.1775, val_r2:0.825
e985, train_loss:0.0757, r2:0.998, val_loss:0.1763, val_r2:0.826
e986, train_loss:0.0842, r2:0.997, val_loss:0.1781, val_r2:0.825
e987, train_loss:0.0869, r2:0.997, val_loss:0.1837, val_r2:0.819
e988, train_loss:0.0763, r2:0.997, val_loss:0.1890, val_r2:0.814
e989, train_loss:0.0844, r2:0.998, val_loss:0.1944, val_r2:0.808
e990, train_loss:0.0825, r2:0.997, val_loss:0.1970, val_r2:0.806
e991, train_loss:0.0901, r2:0.997, val_loss:0.1971, val_r2:0.806
e992, train_loss:0.0851, r2:0.998, val_loss:0.1967, val_r2:0.806
e993, train_loss:0.0751, r2:0.997, val_loss:0.1923, val_r2:0.811
e994, train_loss:0.0870, r2:0.998, val_loss:0.1883, val_r2:0.814
e995, train_loss:0.0795, r2:0.998, val_loss:0.1861, val_r2:0.817
e996, train_loss:0.0837, r2:0.998, val_loss:0.1833, val_r2:0.819
e997, train_loss:0.0847, r2:0.998, val_loss:0.1839, val_r2:0.819
e998, train_loss:0.0915, r2:0.998, val_loss:0.1865, val_r2:0.816
e999, train_loss:0.0853, r2:0.997, val_loss:0.1859, val_r2:0.817